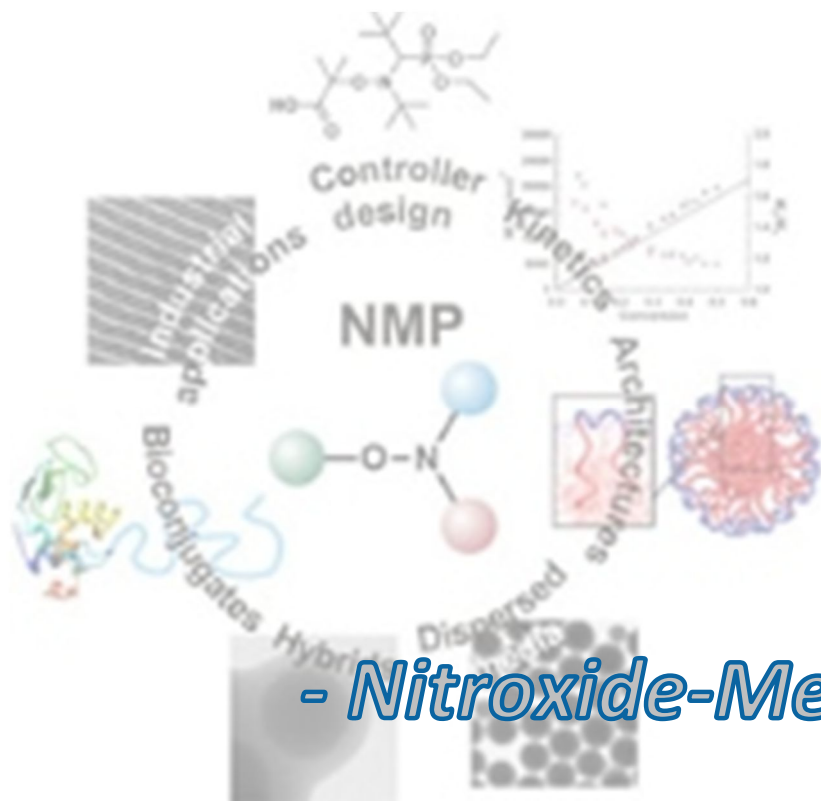




RDRP

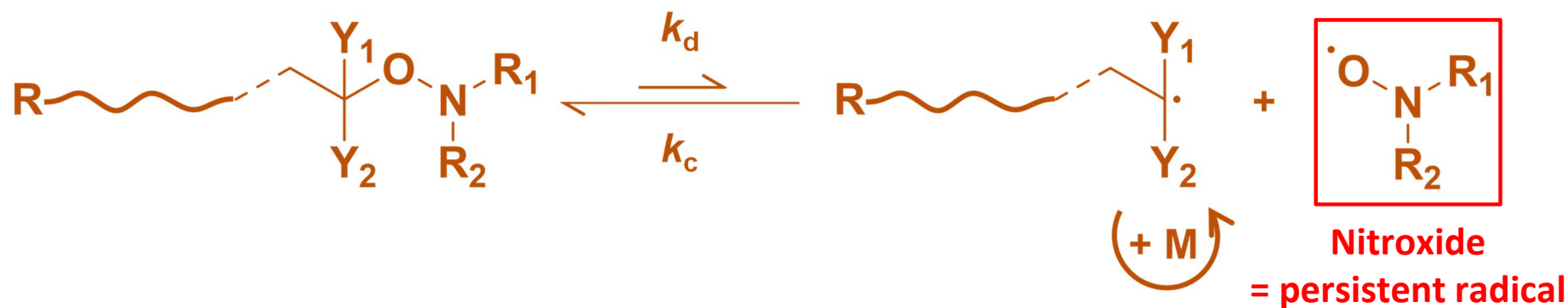
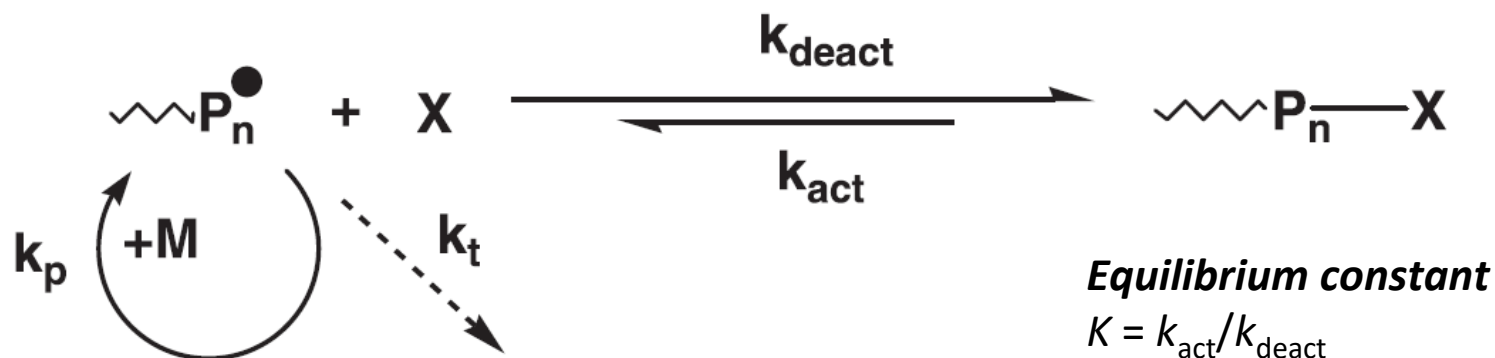
- Nitroxide-Mediated Polymerization -

Prog. Polym. Sci. **2013**, 38, 63–235

Nitroxide-Mediated Polymerization (NMP)

Basic Principles

Nature of the controlling agent X

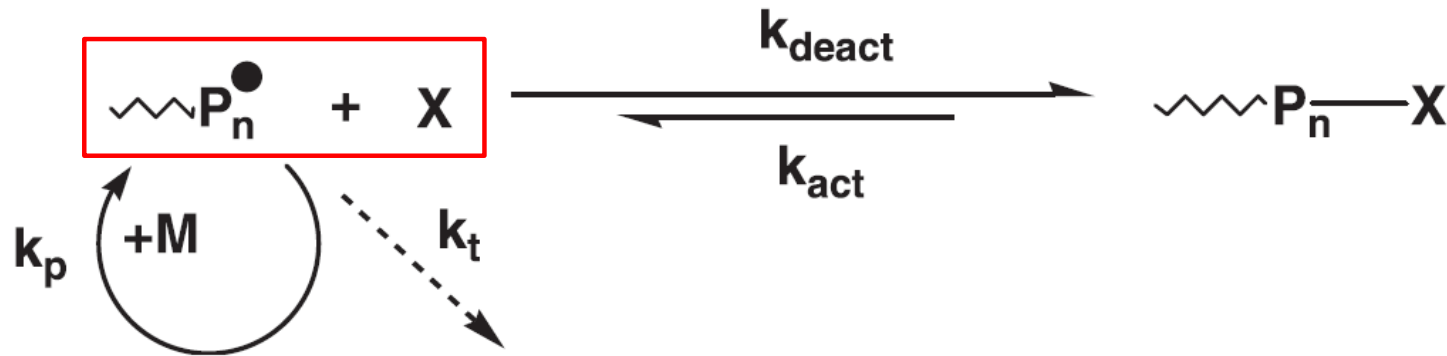


In some (old) cases, **NMP** is also termed SFRP: stable free-radical polymerization.

Nitroxide-Mediated Polymerization (NMP)

Basic Principles

How to start a NMP?



Bicomponent initiating system

We start a classic radical polymerization in the presence of a nitroxide.

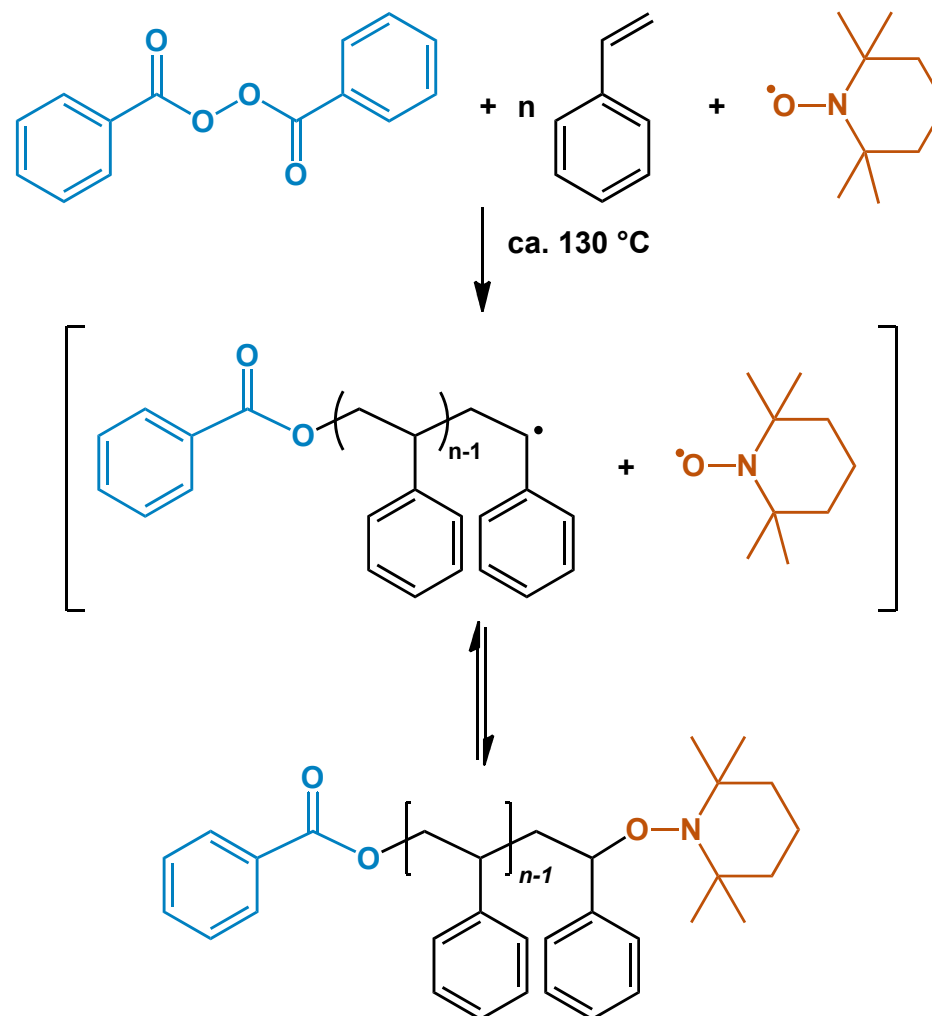
Nitroxide-Mediated Polymerization (NMP)

Basic Principles

How to start a NMP?

The presence of the reversible capping reaction significantly reduces the propagating free-radical concentration, thus minimizing irreversible termination events.

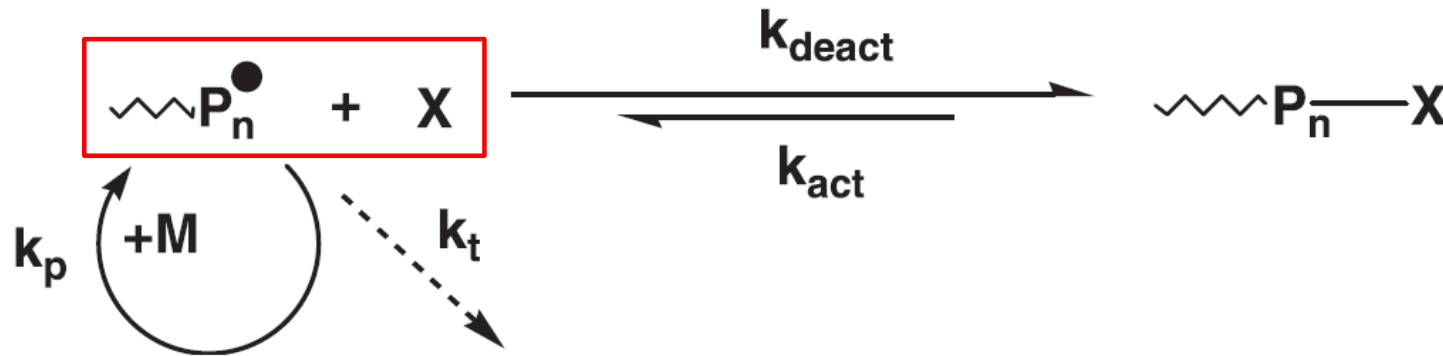
TEMPO-mediated RP at 130 °C



Nitroxide-Mediated Polymerization (NMP)

Basic Principles

How to start a NMP?

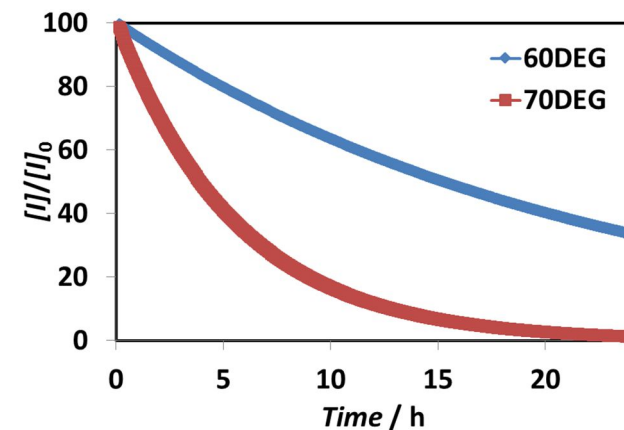


Bicomponent initiating system

We start a classic radical polymerization in the presence of a nitroxide.

Problem: You depend on the initiator decomposition rate and the efficiency...

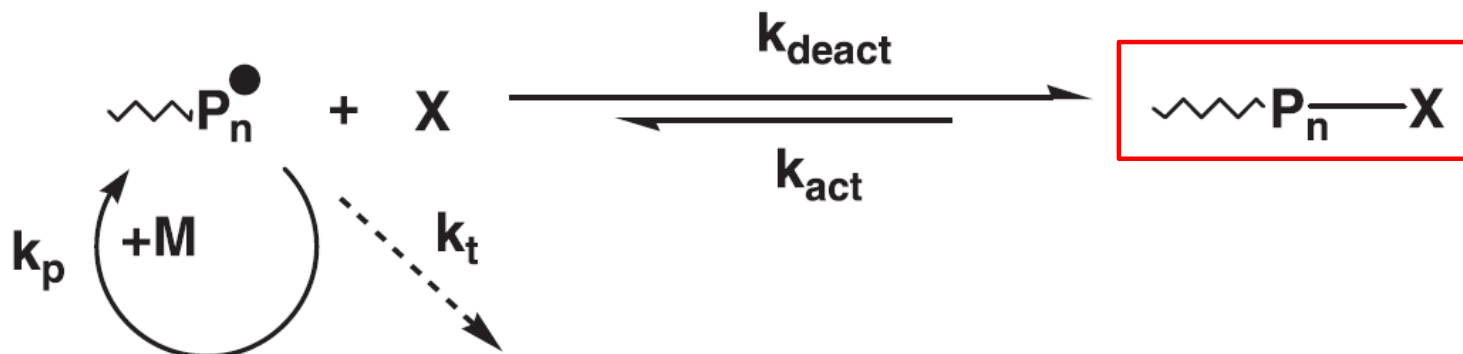
Discrepancy between $[\text{radicals created/initiating}]_t$ and $[\text{nitroxide}]_0$



Nitroxide-Mediated Polymerization (NMP)

Basic Principles

How to start a NMP?



Monocomponent initiating system

We start from an alkoxyamine, which serves as thermal initiator.

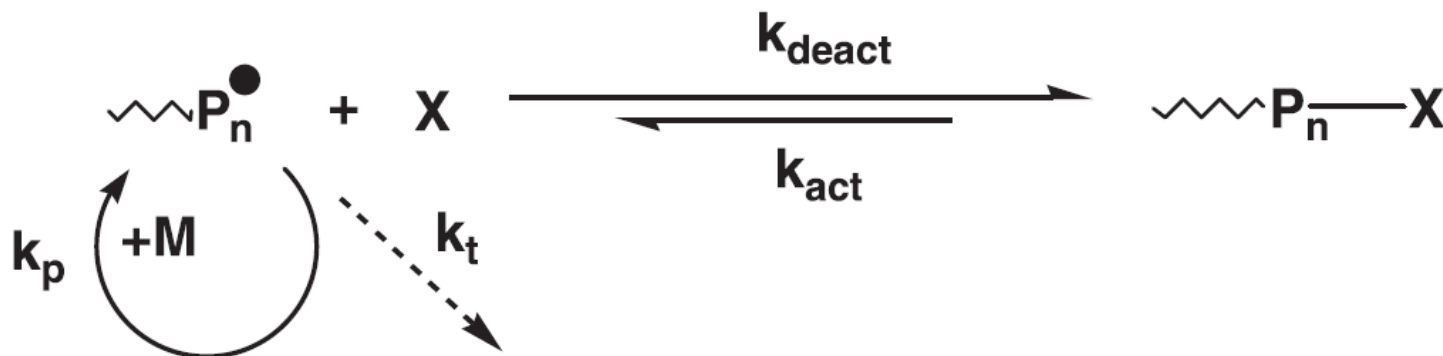
Monomolecular initiating system $\rightarrow$ Arbitrary targeting degrees of polymerization, i.e., molar masses!

$$DP_n \text{ (full conversion)} = [\text{Monomer}]/[\text{Alkoxyamine}]$$

Nitroxide-Mediated Polymerization (NMP)

Basic Principles

The persistent radical effect (PRE)*

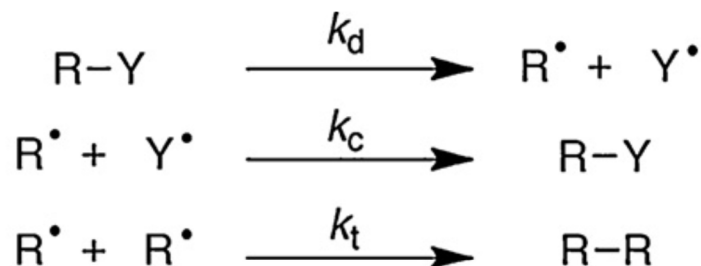


*applies also to ATRP

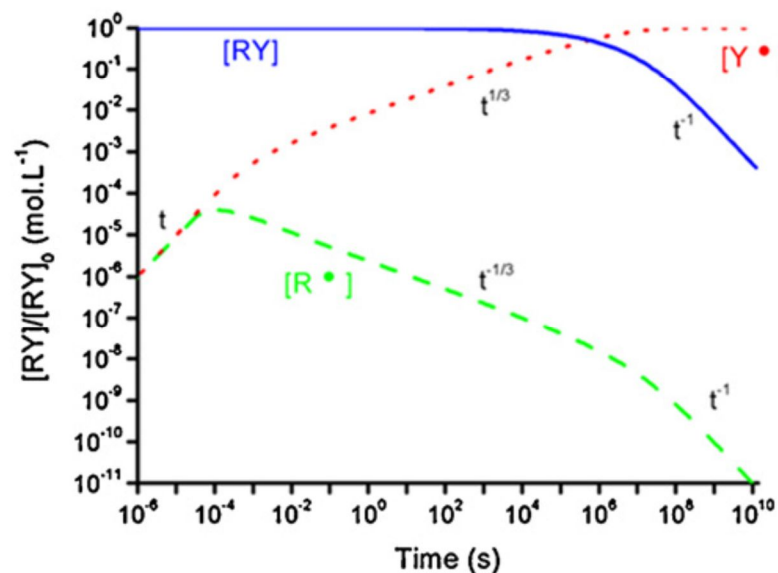
Nitroxide-Mediated Polymerization (NMP)

Basic Principles

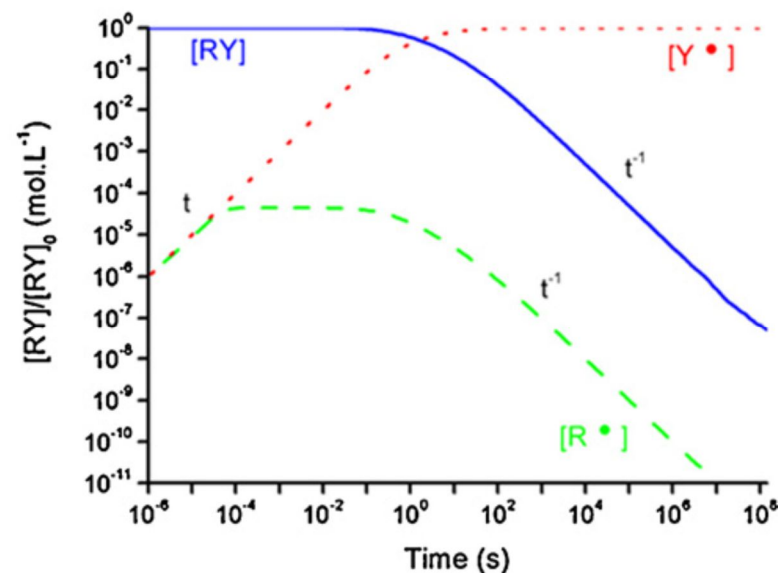
The persistent radical effect (PRE)



(a)



(b)

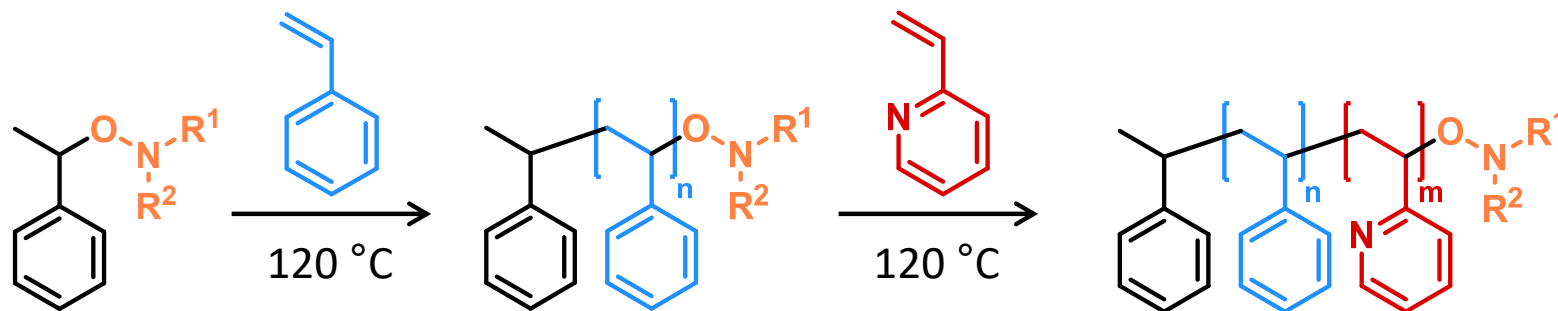


Concentrations of RY , $\text{Y}^\bullet$ and $\text{R}^\bullet$ vs time in a double logarithmic plot with $[\text{RY}]_0 = 5.0 \times 10^{-2} \text{ M}$, $I_d = 10^{-2} \text{ s}^{-1}$, $k_t = 10^8 \text{ M}^{-1} \text{ s}^{-1}$, $k_c = 10^7 \text{ M}^{-1} \text{ s}^{-1}$ (a) and $k_c = 10^4 \text{ M}^{-1} \text{ s}^{-1}$ (b).

Nitroxide-Mediated Polymerization (NMP)

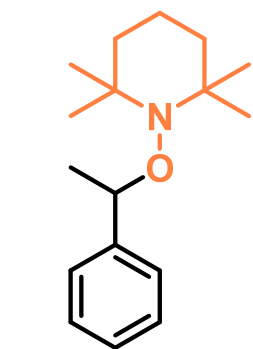
Macromolecular Engineering

Synthesis of block copolymers

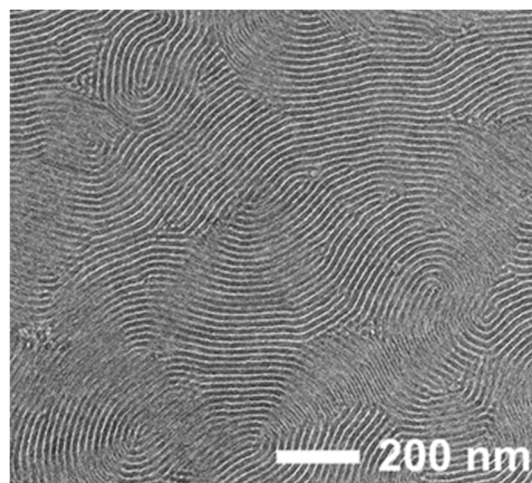


conversion < 70–80 %

Purification by precipitation in methanol



styryl-TEMPO

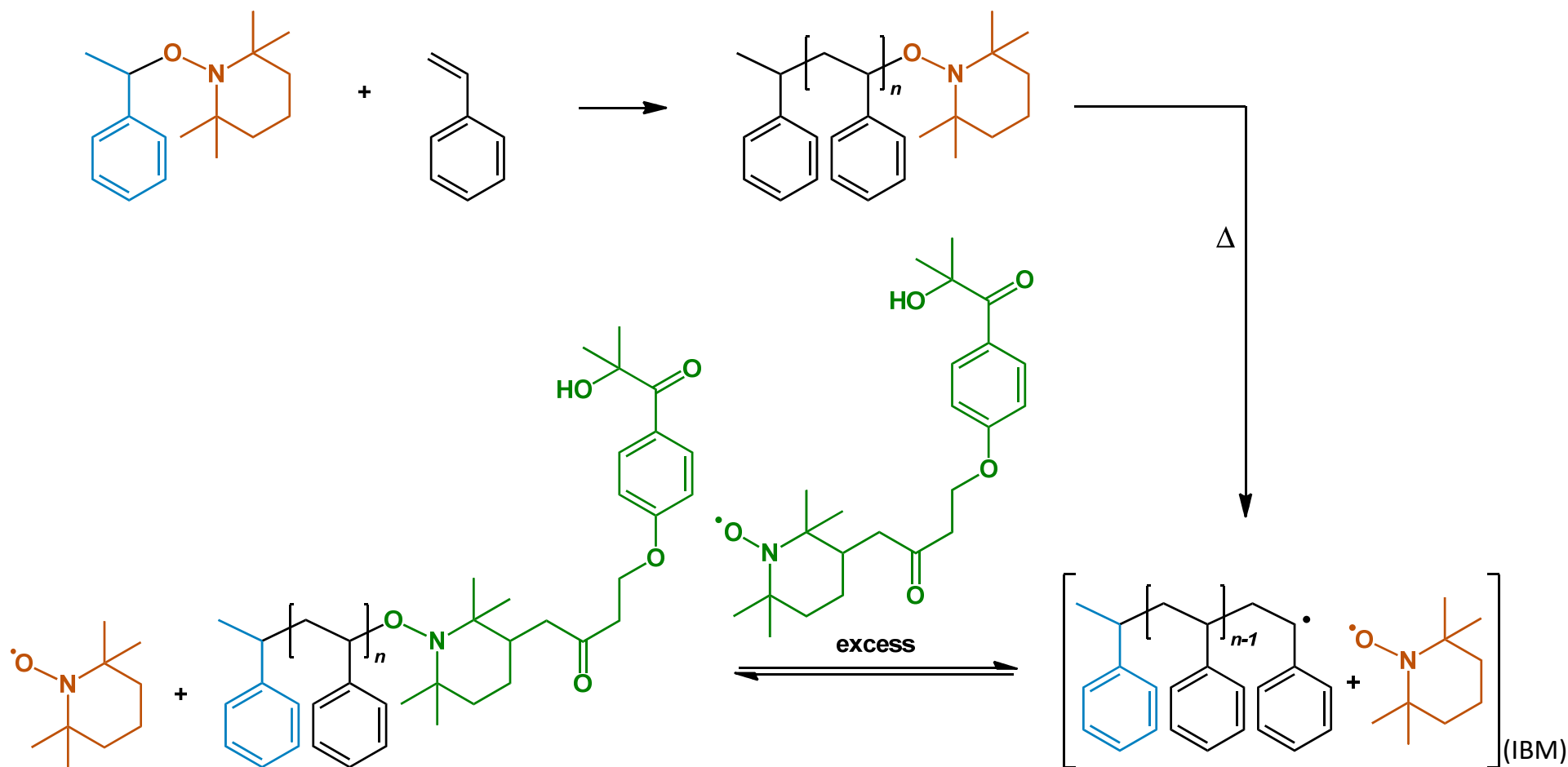


Adv. Mater **2005**, 17, 2591

Nitroxide-Mediated Polymerization (NMP)

Macromolecular Engineering

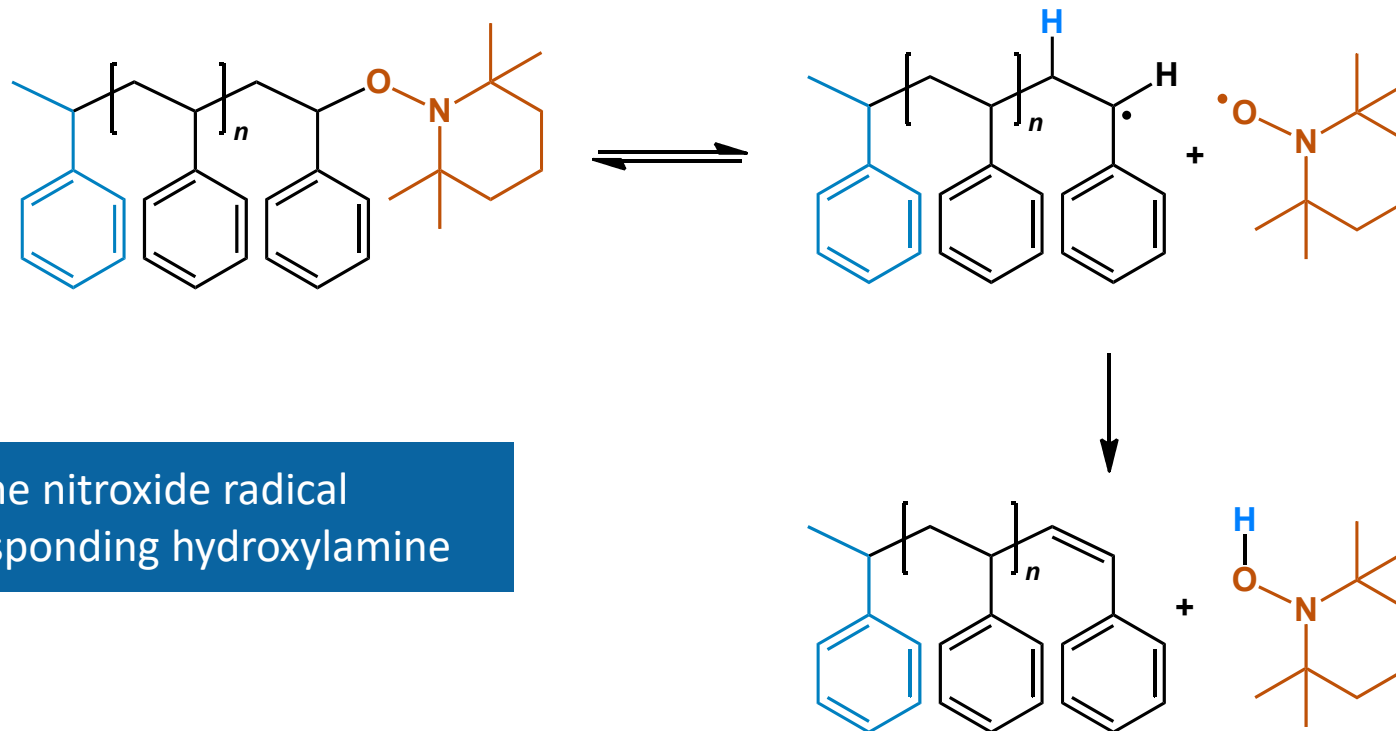
Using NMP for end group functionalization



What can you do with this?

Nitroxide-Mediated Polymerization (NMP)

Potential side reactions



Reduction of the nitroxide radical
leading to the corresponding hydroxylamine

→ Decrease in living character of the polymerization

Nitroxide-Mediated Polymerization (NMP)

Variation of the nitroxide structure

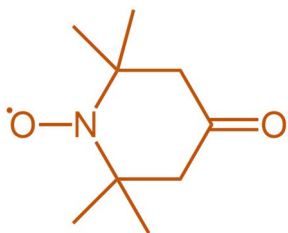
TEMPO was initially the favored nitroxide, but soon serious deficiencies became apparent:

1. Necessity to use high reaction temperatures (i.e., $T > 130\text{ }^{\circ}\text{C}$)
2. Long reaction times ($24 < t < 72\text{ h}$)
3. Incompatibility with a large array of monomers. Living polymerization was only possible with styrene or styrene derivatives.

Change in nitroxide structure



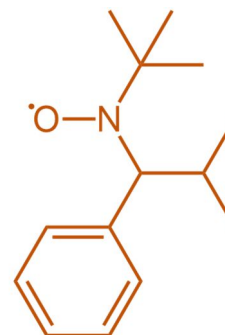
2nd and 3rd generation mediating agents



2nd Generation Nitroxides

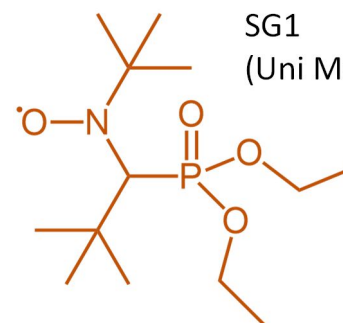
Acrylates at $150\text{ }^{\circ}\text{C}$, $\bar{D} < 1.6$

TIPNO (IBM)



3rd Generation Nitroxides

Acrylates, Acrylamides, $60\text{ }^{\circ}\text{C}$, 1,3-Dienes, $\bar{D} < 1.1$



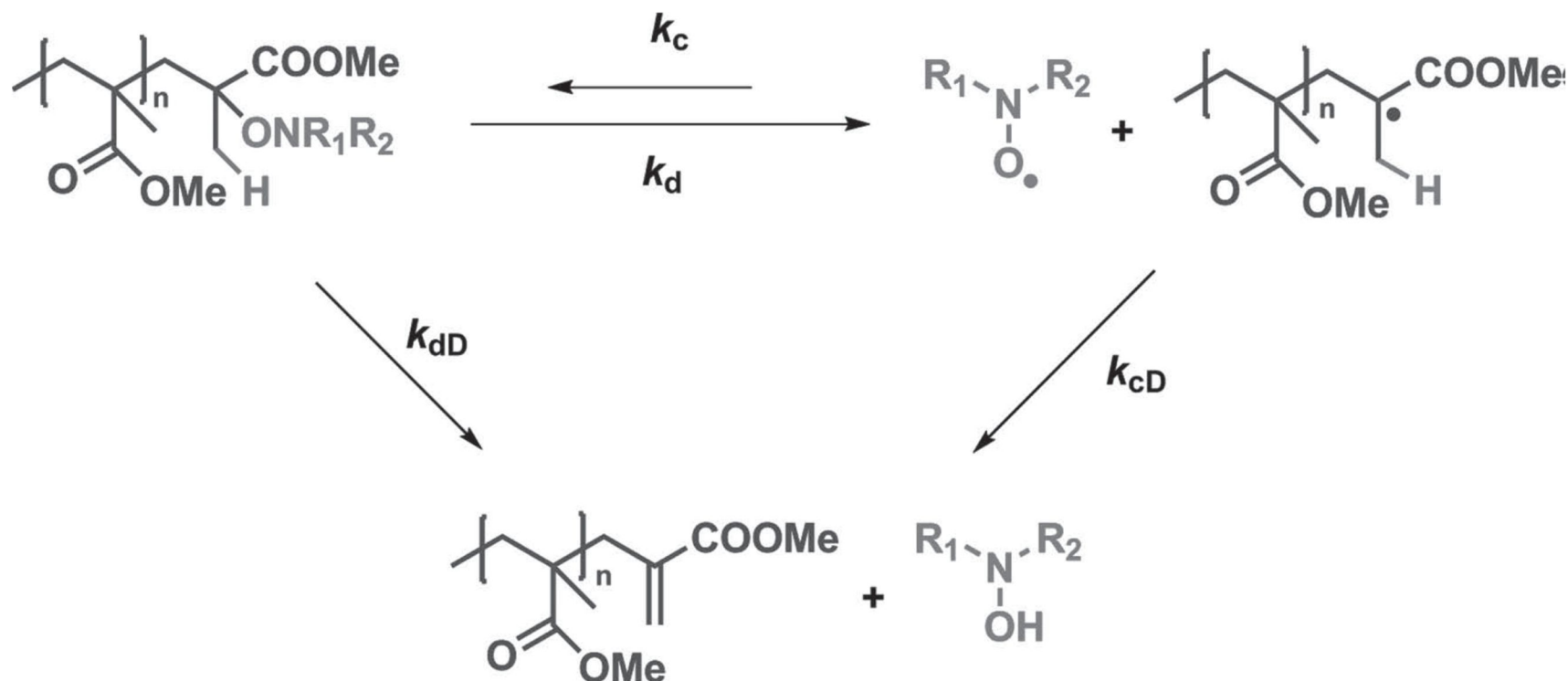
SG1
(Uni Marseille, Arkema)

Nitroxide-Mediated Polymerization (NMP)

Variation of the nitroxide structure

SG1* – a better nitroxide, yet not perfect

Side reaction in the case of methacrylates (elimination and disproportionation)

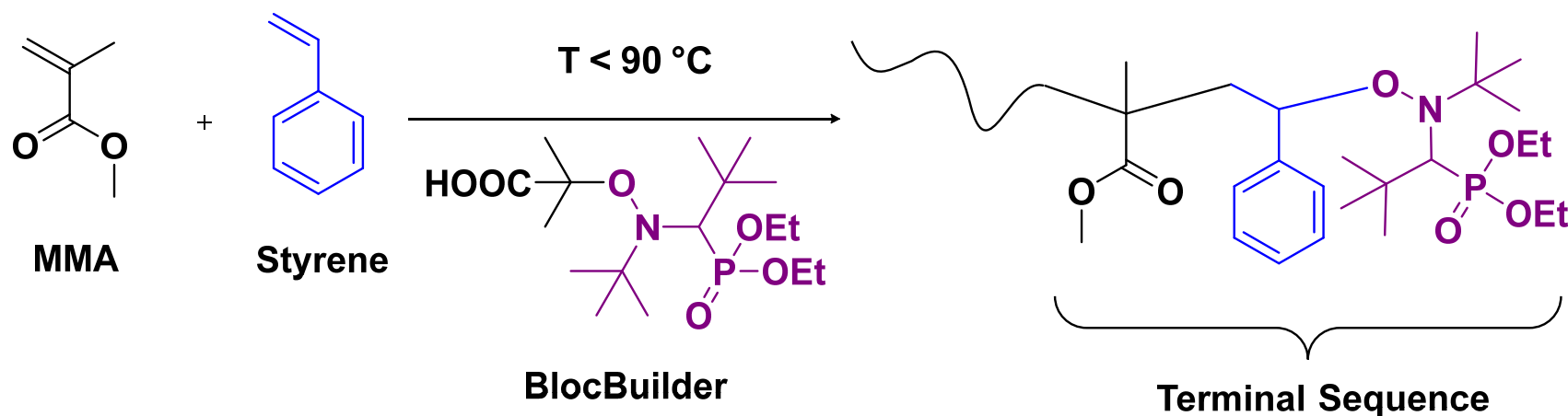


**N*-*tert*-butyl-*N*-[1-diethylphosphono-(2,2-dimethylpropyl)] nitroxide]

Nitroxide-Mediated Polymerization (NMP)

SG1-mediated methacrylate polymerization

The trick!



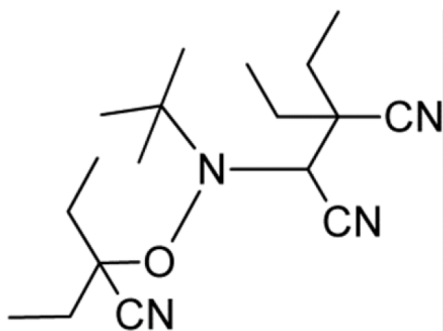
$[\text{styrene}]/[\text{MMA}] \approx 9:91$

→ **Physically** almost like a pure PMMA;

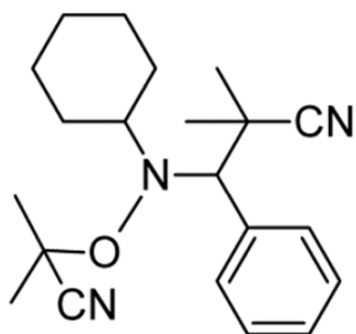
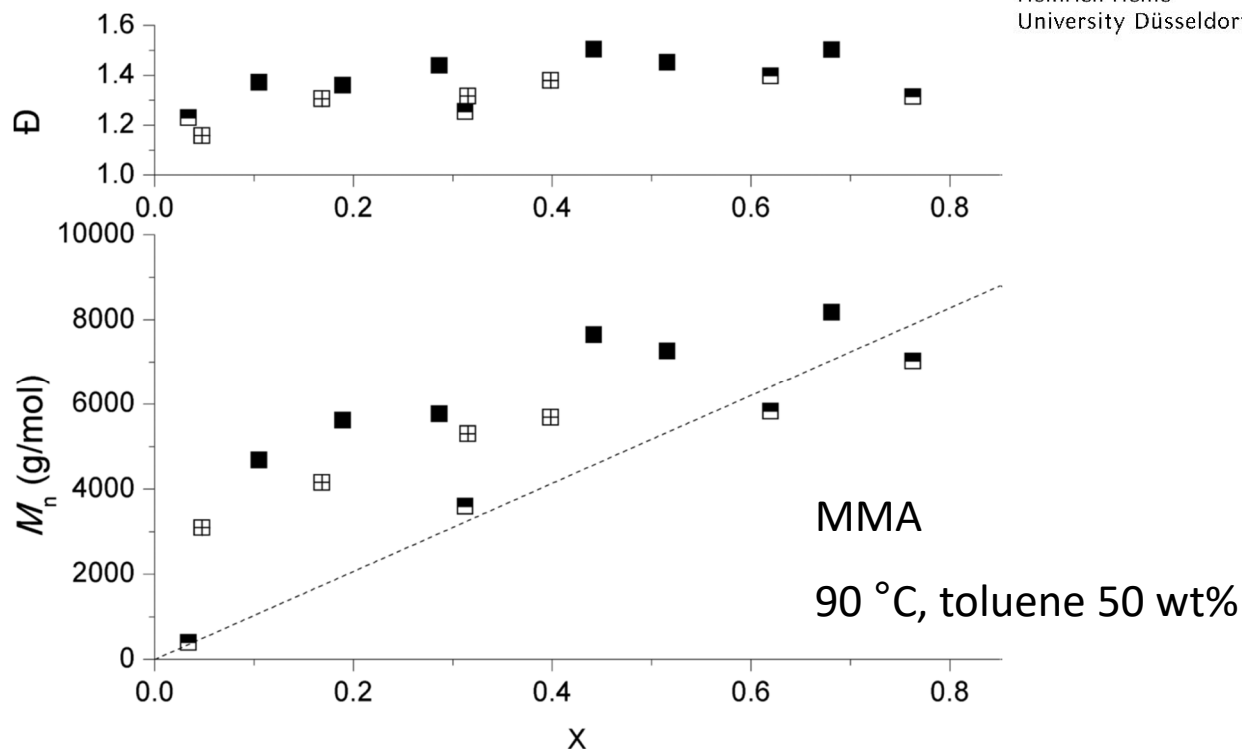
→ **Chemically** almost like a pure PS macroalkoxyamine, except a lower decomposition temperature

Nitroxide-Mediated Polymerization (NMP)

Latest types of alkoxyamines



ACS Macro Lett. **2016**, 5, 1019



Dispolreg 007
(POLYMAT, Uni San Sebastián)

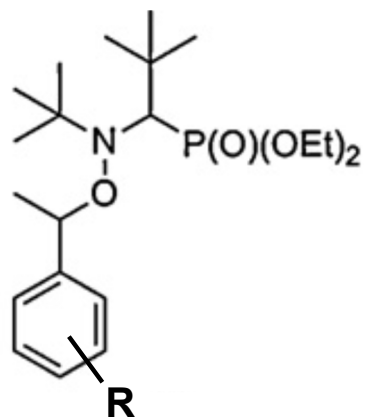
Works for styrene too!

Polym. Chem. **2017**, 8, 1728

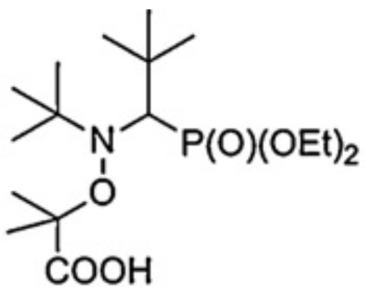
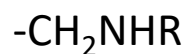
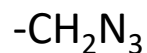
Nitroxide-Mediated Polymerization (NMP)

Macromolecular Engineering

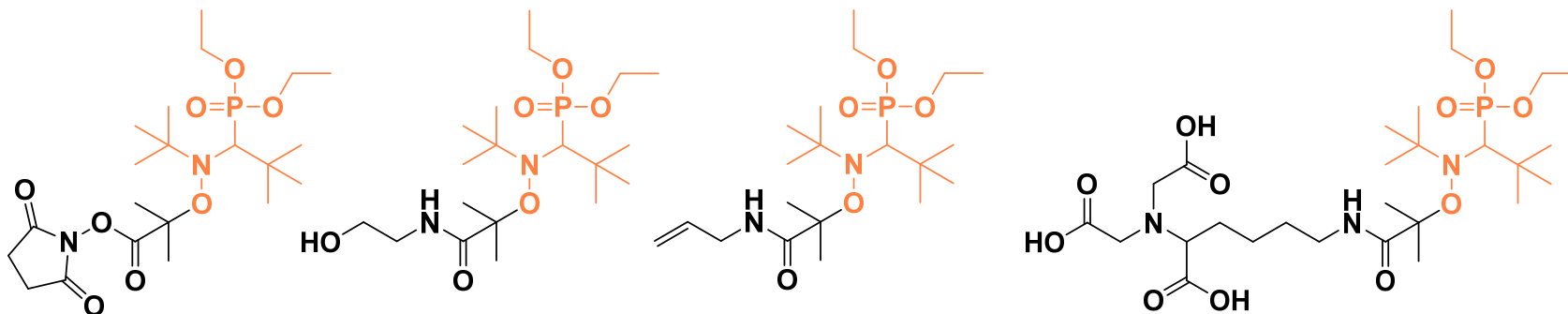
End-group functionalization (α) using custom-made alkoxyamine initiators



Substituents on the phenylethyl fragment



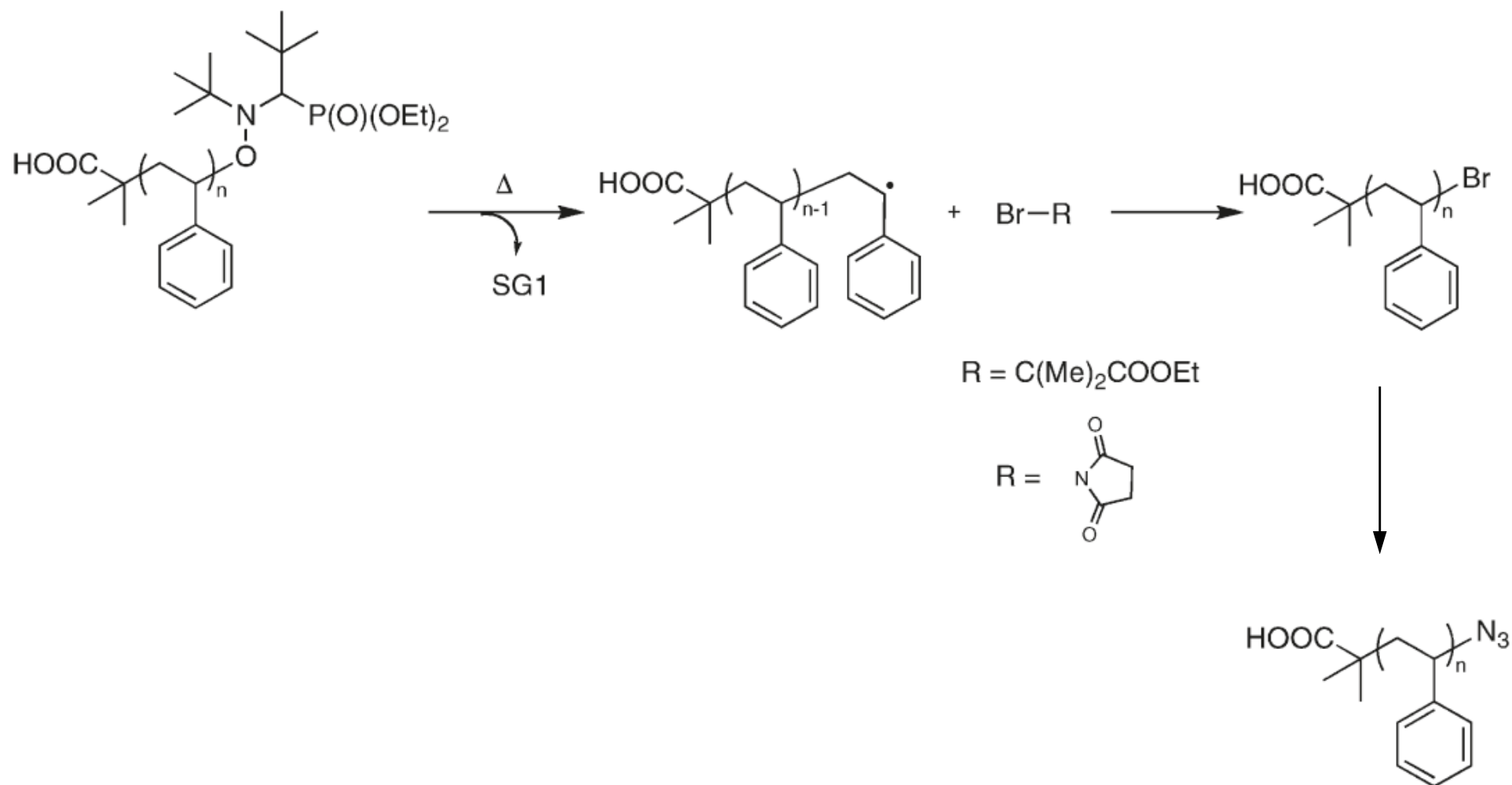
Amide derivatives of BlocBuilder

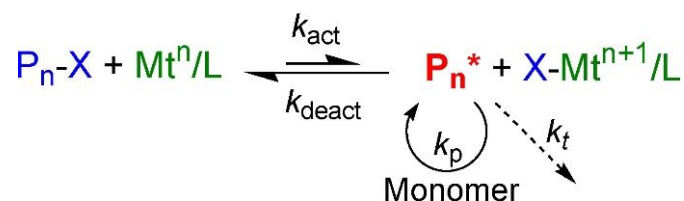
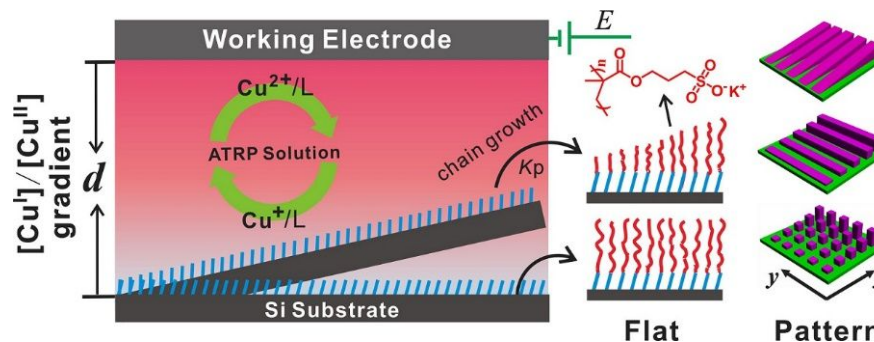


Nitroxide-Mediated Polymerization (NMP)

Macromolecular Engineering

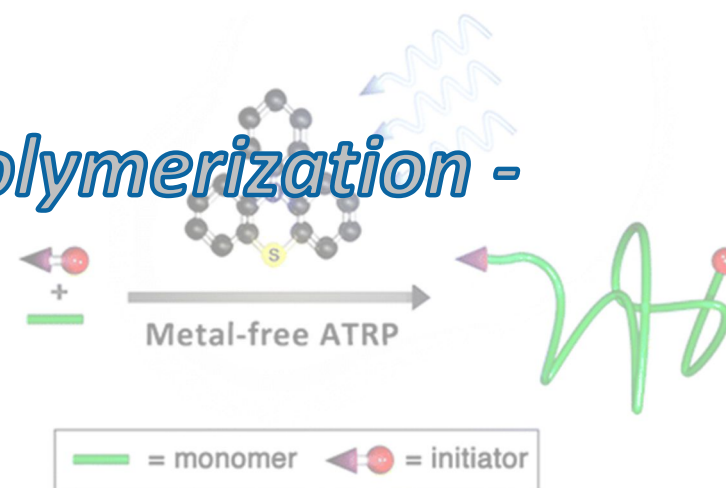
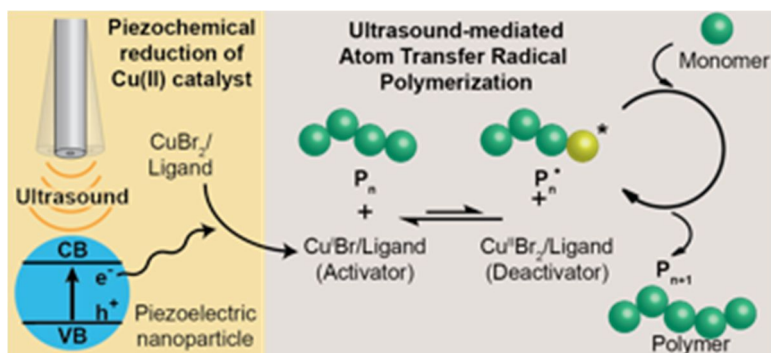
End-group functionalization (ω) $\rightarrow$ End-group transformation





RDRP

- Atom-Transfer Radical Polymerization -

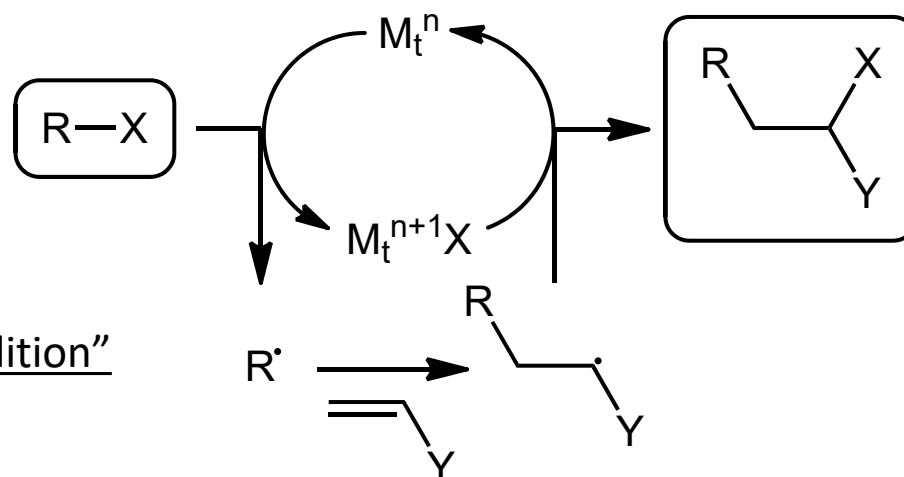


Nat Rev Methods Primers **2025**, 5, 2

Atom-Transfer Radical Polymerization (ATRP)

ATRA - Lowering the activation energy for C-X bond scission by catalysis

Atom-transfer radical addition (**ATRA**) is a well-known method for carbon-carbon bond formation.



Also termed “Kharasch addition”

The transition-metal species, M_t^n , abstracts the halogen atom from the organic halide, $R-X$, to form the oxidized organo-metallic species, $M_t^{n+1}X$, and the carbon centered radical, $R^\bullet$.

In the next step, the radical, $R^\bullet$, reacts with the alkene to give the radical species.

Reaction between the newly formed radical and the $M_t^{n+1}X$ results in the target product and M_t^n .

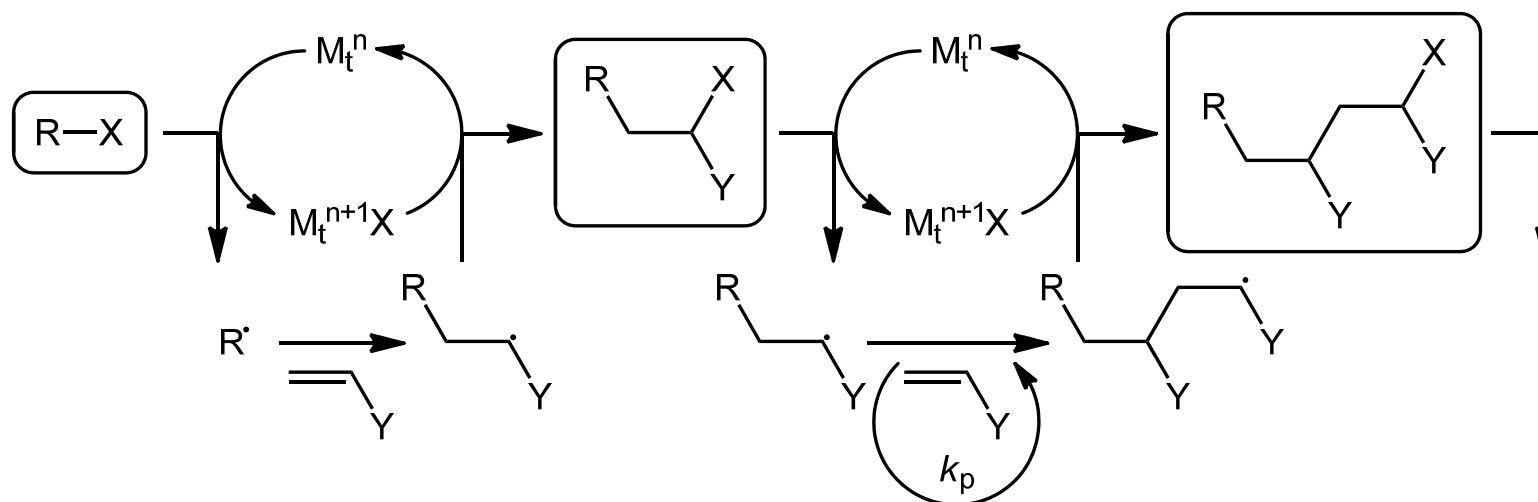
The transition metal-catalyzed atom-transfer reaction is highly efficient in producing the target product. Yields are higher than 90 %.

Atom-Transfer Radical Polymerization (ATRP)

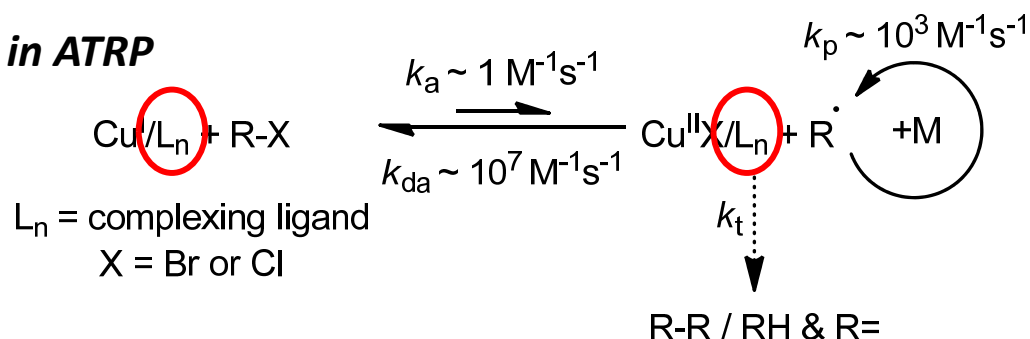
From ATRA to ATRP

The presence of a redox process can effectively reduce the number of radicals, resulting in less significant termination reactions between radicals.

If polymeric halides $R-M_i-X$ are reactive enough towards M_t^n and the monomer is present in excess, repetitive atom-transfer reactions are likely to occur.



Typical rate coefficients in ATRP



Atom-Transfer Radical Polymerization (ATRP)

Transition metals used in ATRP

1

H

1.008

2

He

4.003

3

Li

6.941

4

Be

9.012

11

Na

22.990

12

Mg

24.305

5

B

10.811

6

C

12.011

7

N

14.007

8

O

15.999

9

F

18.998

10

Ne

20.180

13

Al

26.982

14

Si

28.086

15

P

30.974

16

S

32.066

17

Cl

35.453

18

Ar

39.948

19

K

39.098

20

Ca

40.078

21

Sc

44.956

22

Ti

47.88

23

V

50.942

24

Cr

51.996

25

Mn

54.93

26

Fe

55.847

27

Co

58.933

28

Ni

58.69

29

Cu

63.546

30

Zn

65.39

31

Ga

69.723

32

Ge

72.61

33

As

74.922

34

Se

78.96

35

Br

79.904

36

Kr

83.80

37

Rb

85.468

38

Sr

87.62

39

Y

88.906

40

Zr

91.224

41

Nb

92.906

42

Mo

95.94

43

Tc

(98)

44

Ru

101.07

45

Rh

102.906

46

Pd

106.42

47

Ag

107.868

48

Cd

112.411

49

In

114.82

50

Sn

118.72

51

Sb

121.75

52

Te

127.60

53

I

126.906

54

Xe

131.29

55

Cs

132.905

56

Ba

137.327

71

Lu

174.967

72

Hf

178.49

73

Ta

180.948

74

W

183.85

75

Re

186.207

76

Os

190.2

77

Ir

192.22

78

Pt

195.08

79

Au

196.967

80

Hg

200.59

81

Tl

204.383

82

Pb

207.2

83

Bi

209.980

84

Po

(209)

85

At

(210)

86

Rn

(222)

87

Fr

(223)

88

Ra

226.025

103

Lr

(262)

104

(261)

105

(262)

106

(263)

107

(262)

108

(265)

109

(268)

110

(269)

111

(272)

112

(277)

57

La

138.906

58

Ce

140.115

59

Pr

140.908

60

Nd

144.24

61

Pm

(145)

62

Sm

150.36

63

Eu

151.965

64

Gd

157.25

65

Tb

158.925

66

Dy

162.50

67

Ho

164.93

68

Er

167.26

69

Tm

168.934

70

Yb

173.04

89

Ac

227.028

90

Th

232.038

91

Pa

231.036

92

U

238.029

93

Np

237.048

94

Pu

(244)

95

Am

(243)

96

Cm

(247)

97

Bk

(247)

98

Cf

(251)

99

Es

(252)

100

Fm

(257)

101

Md

(260)

102

No

(259)

Periodic Table of the Elements

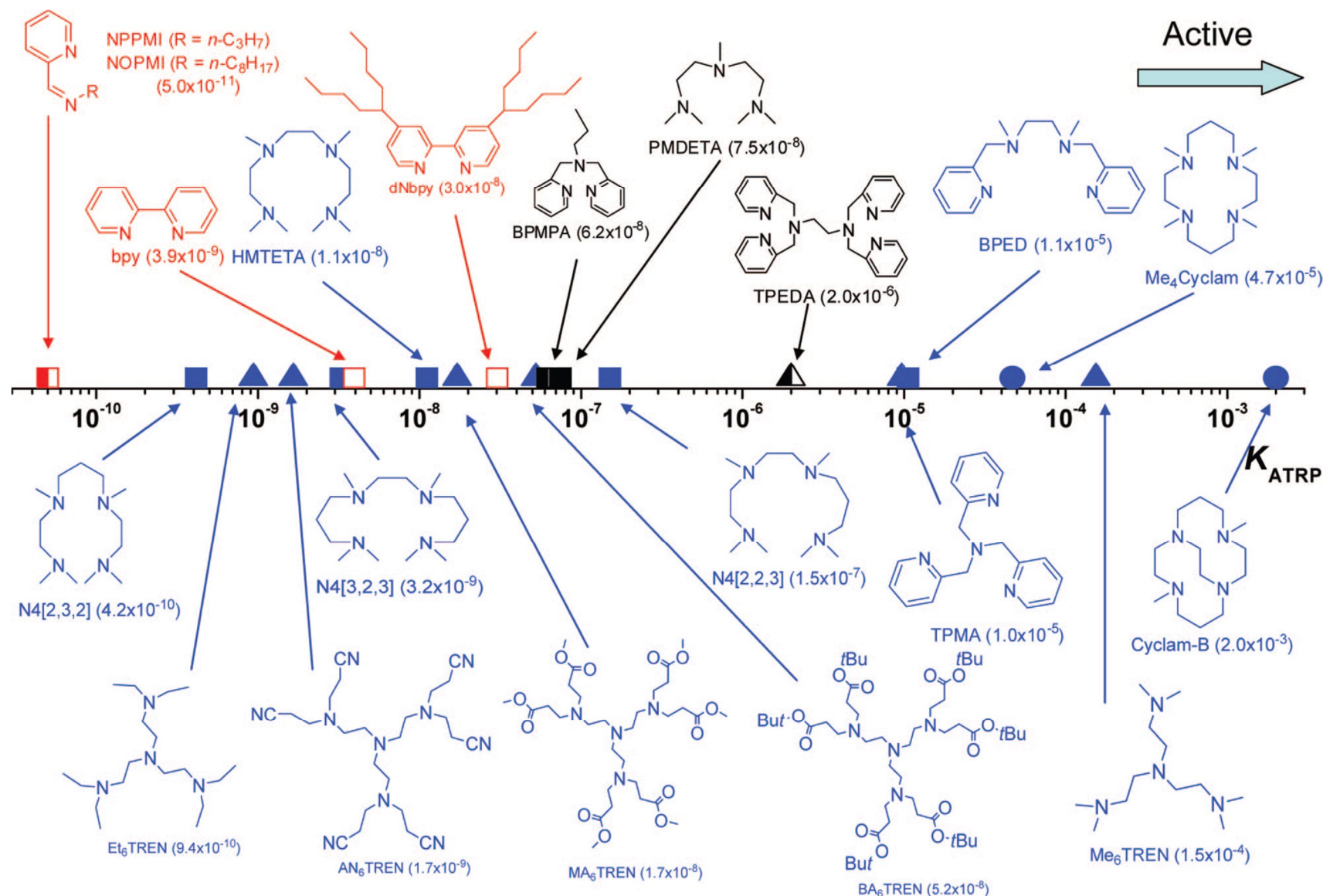
**Mitsuo
Sawamoto**



**Krzysztof
Matyjaszewski**

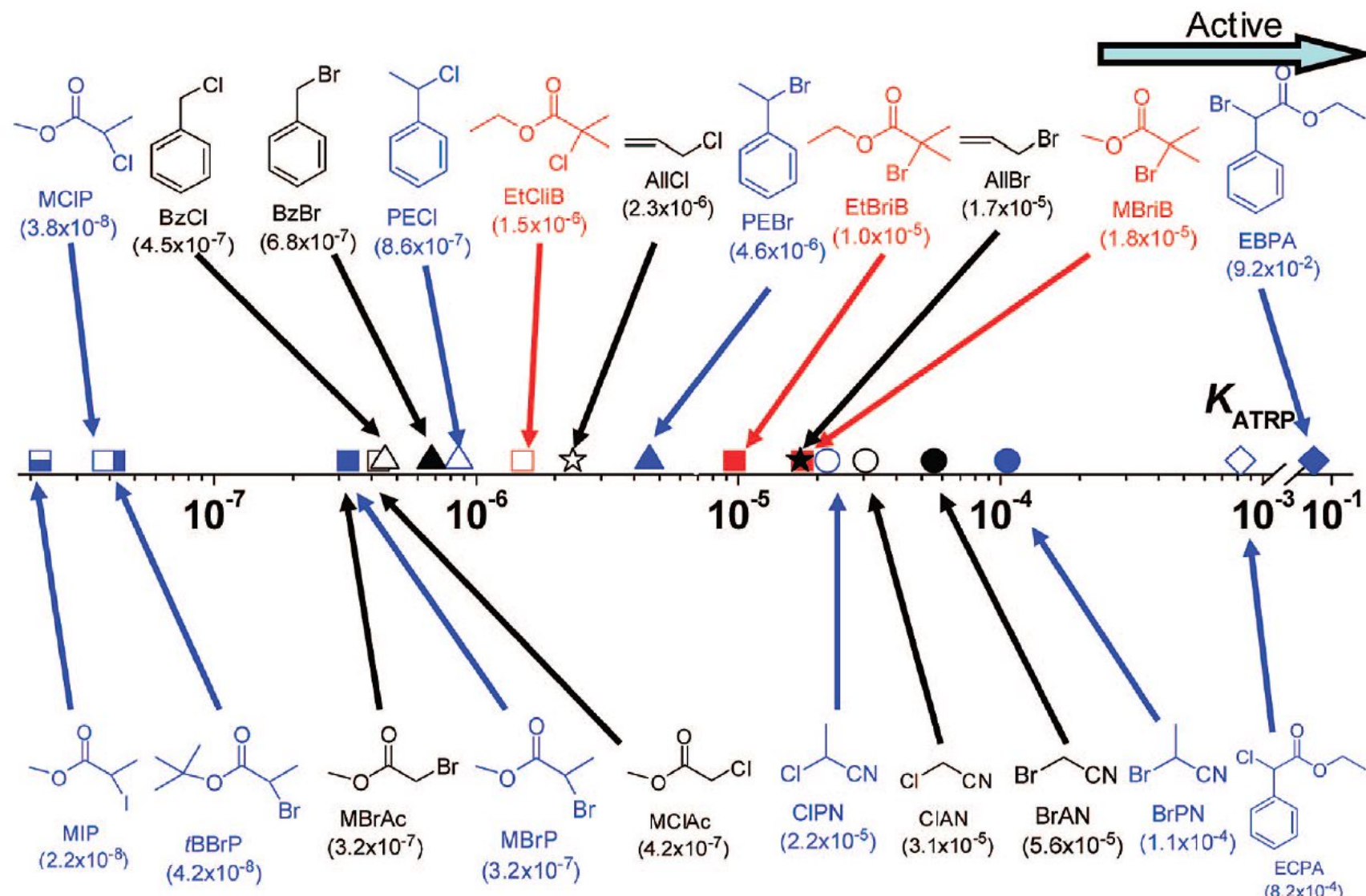
Atom-Transfer Radical Polymerization (ATRP)

Ligands used in ATRP



Atom-Transfer Radical Polymerization (ATRP)

Initiators used in ATRP



Atom-Transfer Radical Polymerization (ATRP)

Typical “old” example



Typical reaction set-up for performing an ATRP experiment

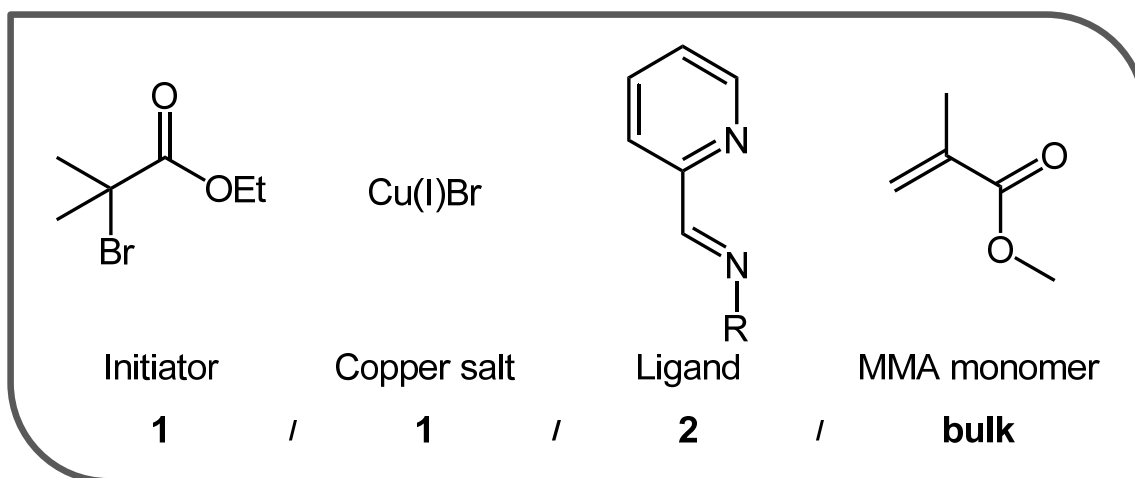


Final purified (chromatography) polymer

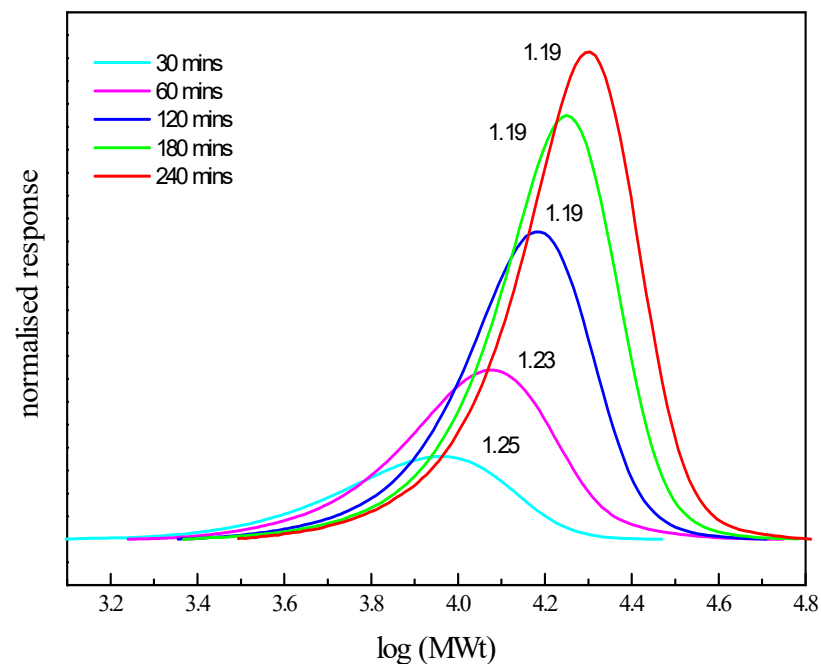
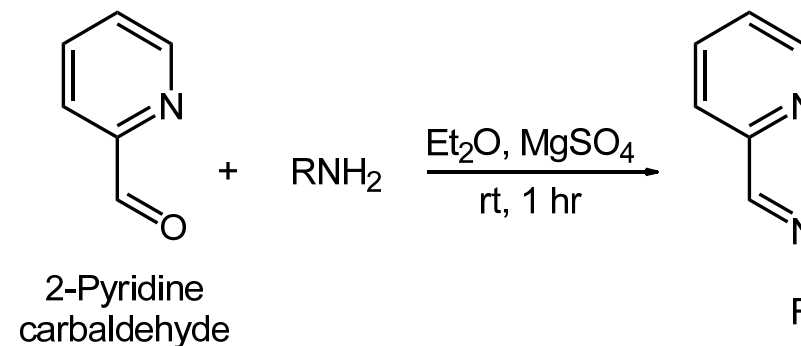
(Warwick)

Atom-Transfer Radical Polymerization (ATRP)

Typical “old” example



Synthesis of the ligand:



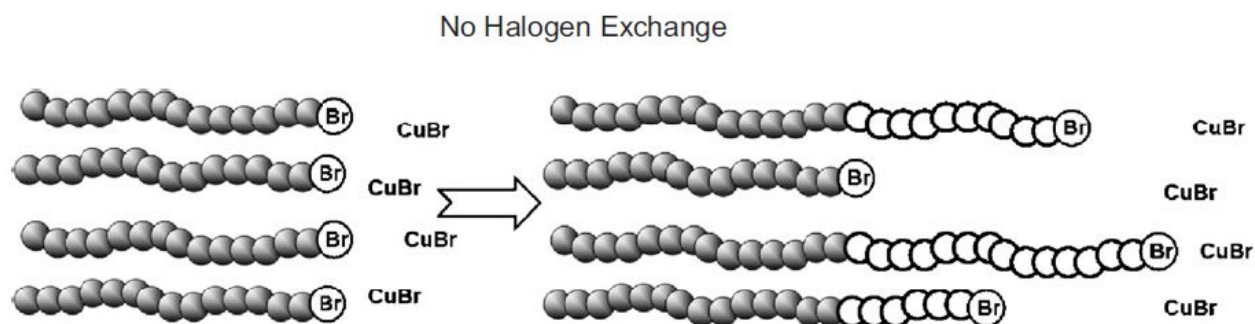
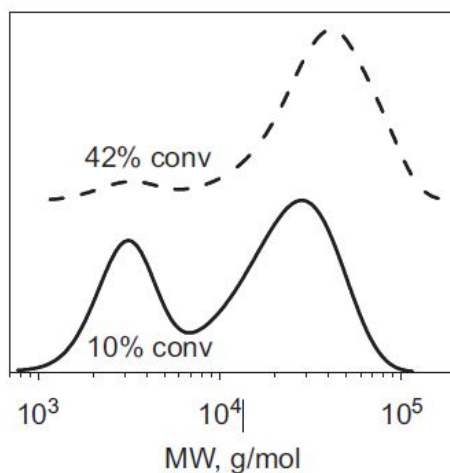
(Warwick)

Atom-Transfer Radical Polymerization (ATRP)

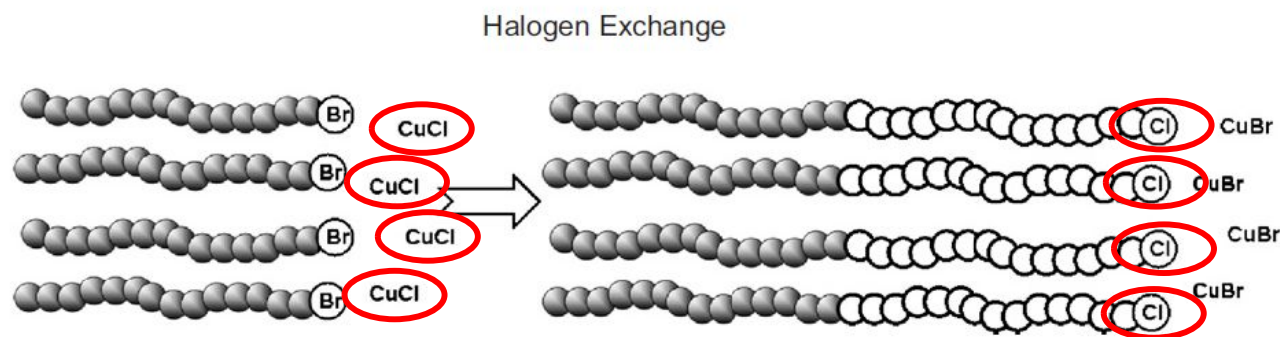
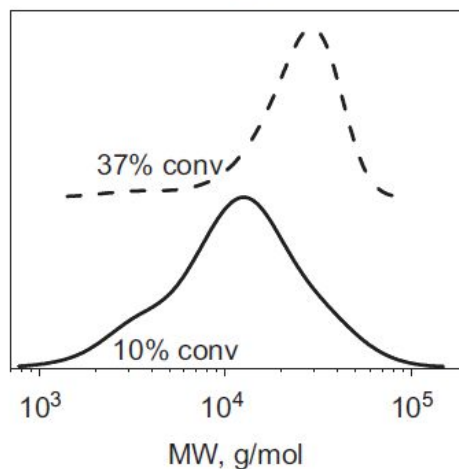
Block copolymer formation – importance of monomer sequence

If I want to make a block copolymer with a PMMA segment and a PS segment, in which order should I proceed? **PS first or PMMA first?**

CuBr / dNbpy + P-Br



CuCl / dNbpy + P-Br

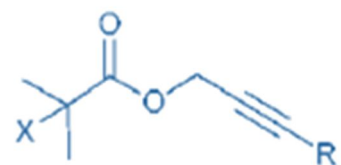


(Carnegie Mellon)

Atom-Transfer Radical Polymerization (ATRP)

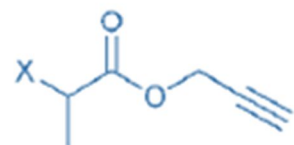
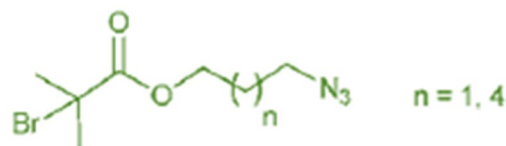
Macromolecular Engineering

End-functional polymers (α): functional initiators

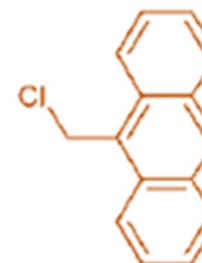
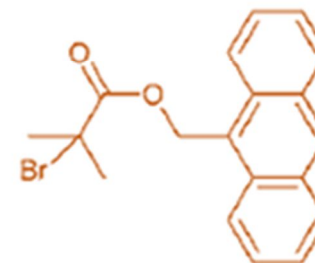
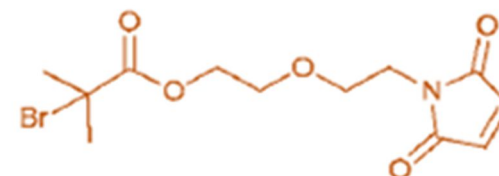
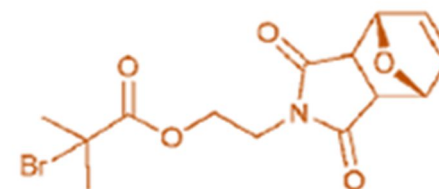
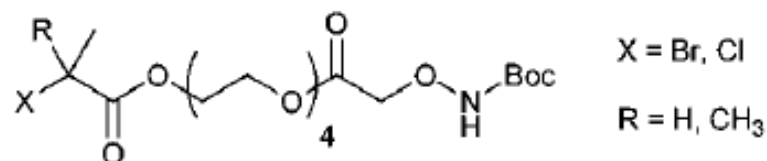
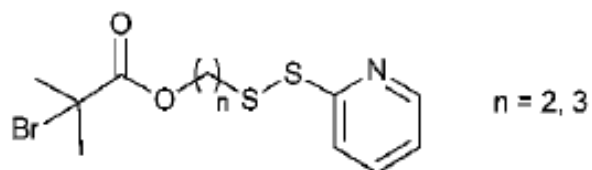
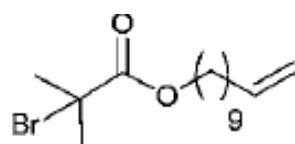
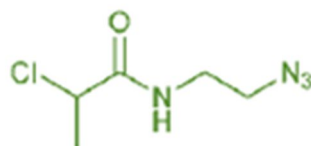
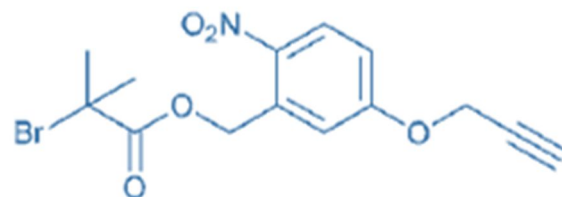
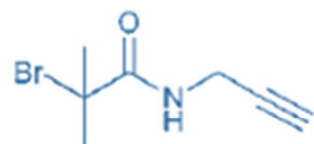


R = TIPS, TMS, H

X = Br, Cl



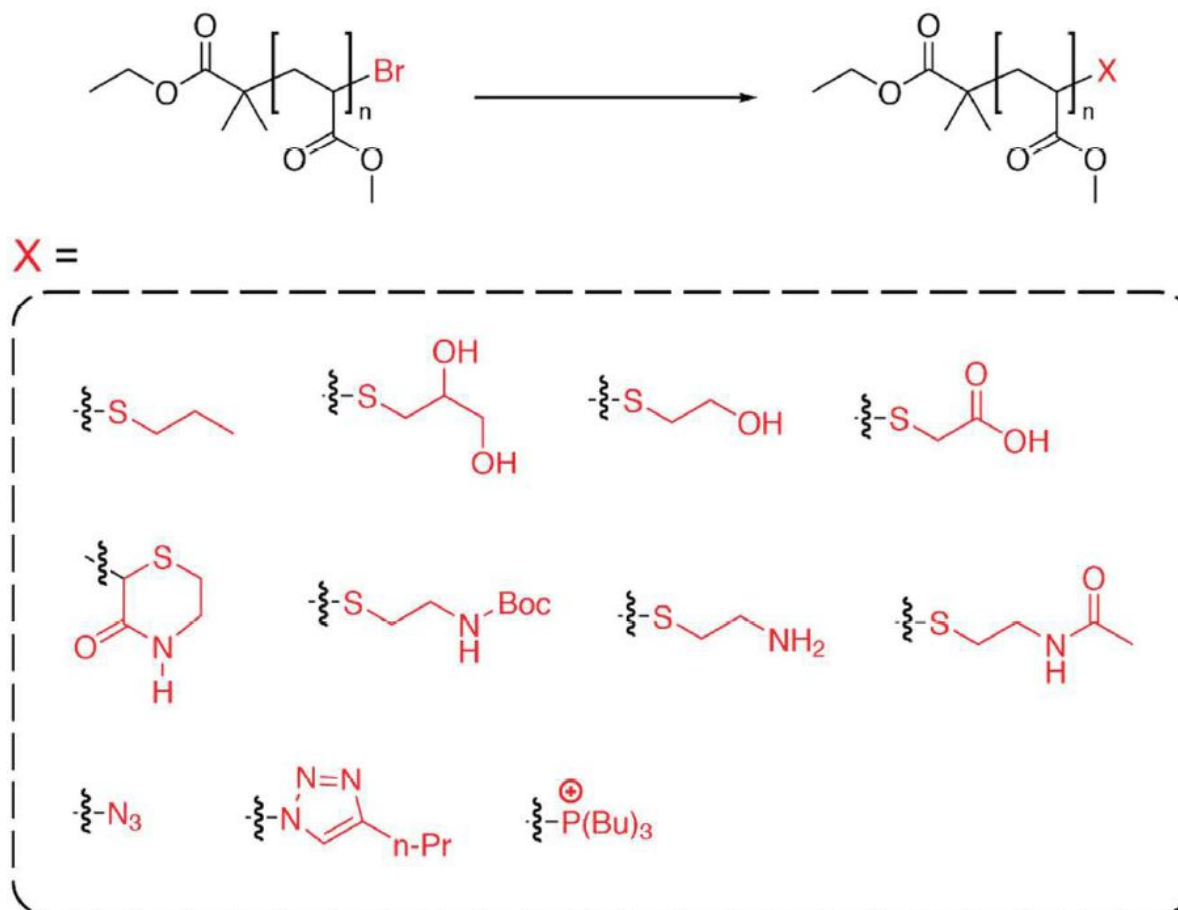
R = Br, Cl



Atom-Transfer Radical Polymerization (ATRP)

Macromolecular Engineering

End-group transformation (S_N2)



Polym. Chem. **2017**, 8, 689

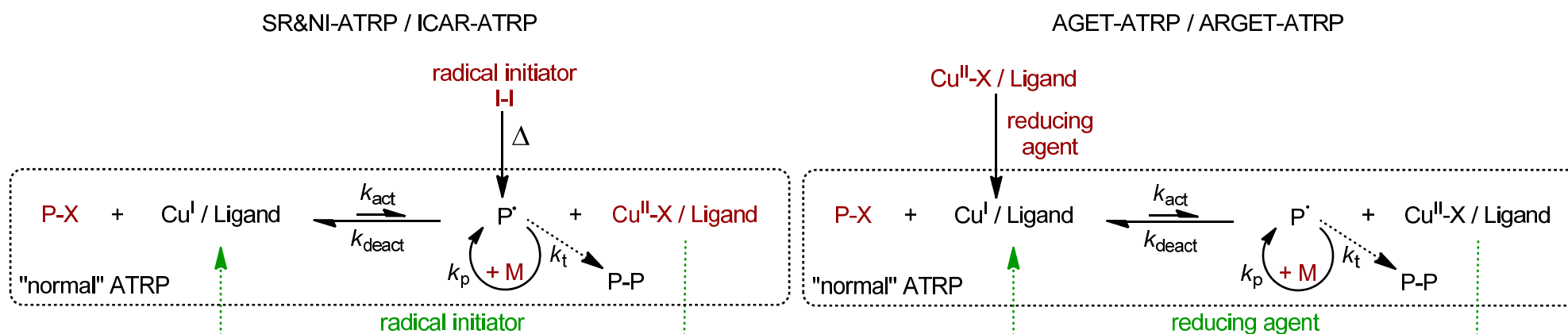
ATRP

Availability of reagents	●
Number of reagents	●
Toxicity	●
Coloration	●
Functional group tolerance	●
Temperature conditions	●
Solvent conditions	●
Sensitivity to oxygen	●
Sensitivity to light	●
Heterogeneous processes	●
End-functional polymers	●
Telechelic polymers	●

Reducing sensitivity to oxygen

The main drawback of „normal“ ATRP is the employment of air-sensitive Cu(I) species. Especially to achieve high polymerization rates, the copper complexes need to be very reactive which makes it even harder to handle them.

Advanced ATRP protocols thus employ the more stable Cu(II) salts in combination with a reducing agent.



The advanced ATRP protocols differ according to the type of reducing agent:

- **Simultaneous reverse and normal initiated ATRP (SN&NI-ATRP):**
radical initiator as the reducing agent, e.g. AIBN
- **Activator generated by electron transfer ATRP (AGET-ATRP):**
reducing agent that does not form radicals, e.g., tin(II)-2-ethylhexanoate, ascorbic acid, MAO.

$$R_p = k_p[M][P_n^*] = k_p K_{\text{ATRP}} \left(\frac{[P_n X][Cu^I/L][M]}{[X-Cu^{II}/L]} \right)$$

$$\frac{M_w}{M_n} = 1 + \frac{1}{DP_n} + \left(\frac{k_p[P_n X]}{k_{\text{deact}}[X-Cu^{II}/L]} \right) \left(\frac{2}{p} - 1 \right)$$

Reducing catalyst amounts

Employing highly active copper complexes with high deactivation rates (k_{deact}), the copper concentration can be minimized to ~ 50 ppm. The Cu^{II} species are regenerated by the additional reducing agent:

- **Activator regenerated by electron transfer ATRP (ARGET-ATRP):**
reducing agent does not form radicals
- **Initiators for continuous activator regeneration (ICAR-ATRP):**
radical initiator as reducing agent.

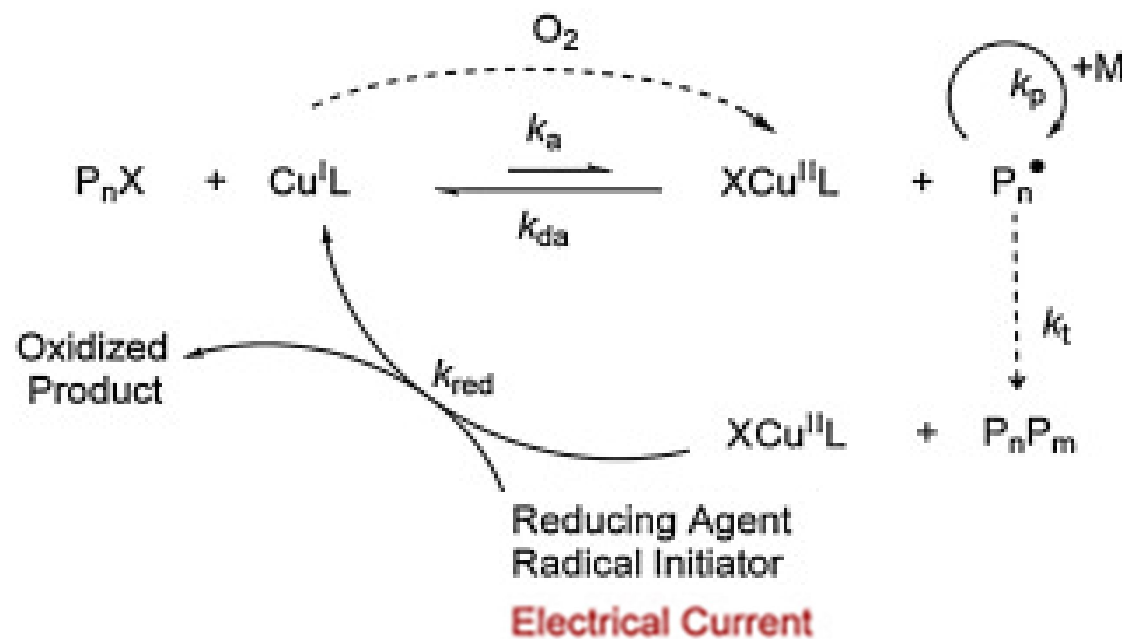
Protocol	Initial reaction mixture
ATRP ¹	$[\text{M}] / [\text{Cu}^{\text{I}}/\text{L}] / [\text{I}] \approx 350 / 1 / 0.3$
SR&NI ATRP ¹	$[\text{M}] / [\text{Cu}^{\text{II}}/\text{L}] / [\text{I}] / [\text{Rad}] \approx 350 / 1 / 0.3 / 0.5$
AGET ATRP ¹	$[\text{M}] / [\text{Cu}^{\text{II}}/\text{L}] / [\text{I}] / [\text{Red}] \approx 350 / 1 / 0.3 / 0.5$
ARGET ATRP ²	$[\text{M}] / [\text{Cu}^{\text{II}}/\text{L}] / [\text{I}] / [\text{Red}] \approx 20\,000 / 1 / 100 / 0.1$
ICAR ATRP ²	$[\text{M}] / [\text{Cu}^{\text{II}}/\text{L}] / [\text{I}] / [\text{Rad}] \approx 20\,000 / 1 / 100 / 10$
[M] monomer; $[\text{Cu}^{\text{I}}/\text{L}]$ copper (I) salt and respective ligand; $[\text{Cu}^{\text{II}}/\text{L}]$ copper (II) salt and respective ligand; [I] initiator (pseudo)halide; [Rad] reducing agent forming radicals; [Red] non-radical reducing agent	

¹Macromolecules **2005**, 38, 4139-4146; ²Proc. Nat. Acad. Sci. USA **2006**, 103, 15309-15314.

	ATRP	Adv. ATRP
Availability of reagents	●	●
Number of reagents	●	●
Toxicity	●	●
Coloration	●	●
Functional group tolerance	●	●
Temperature conditions	●	●
Solvent conditions	●	●
Sensitivity to oxygen	●	●
Sensitivity to light	●	●
Heterogeneous processes	●	●
End-functional polymers	●	●
Telechelic polymers	●	●

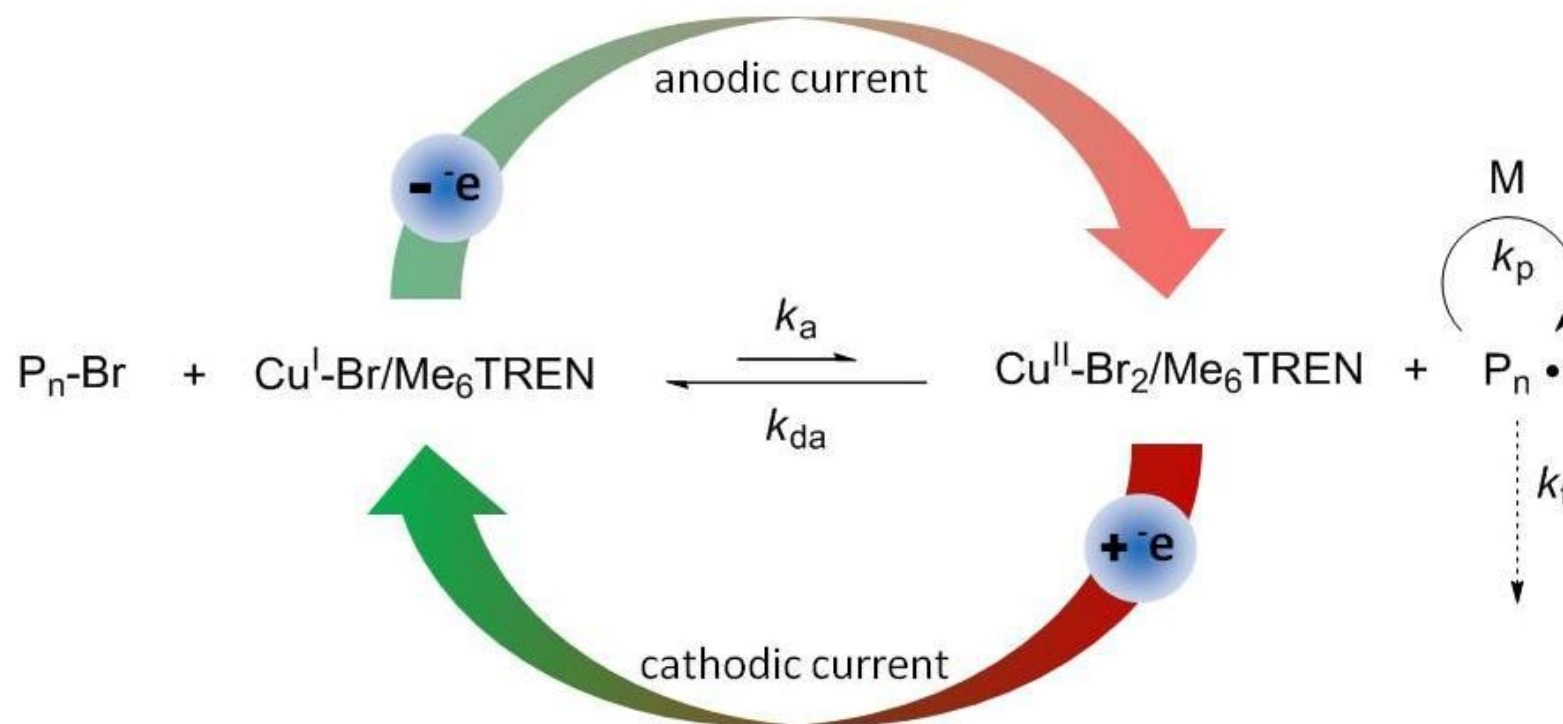
Inducing ATRP by external triggers

Electrochemical control: eATRP



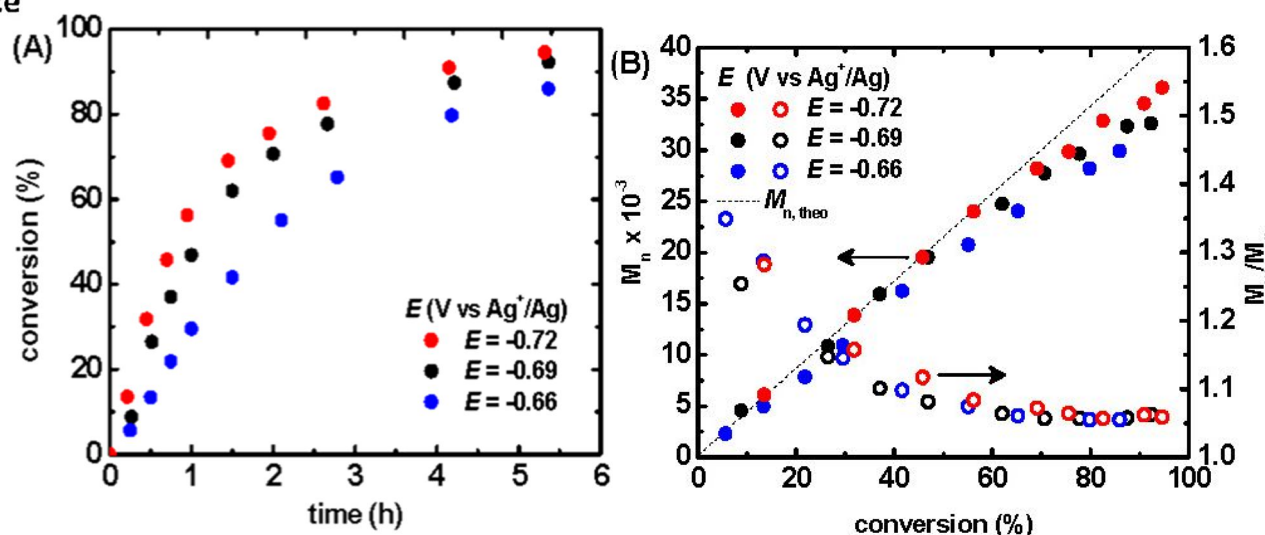
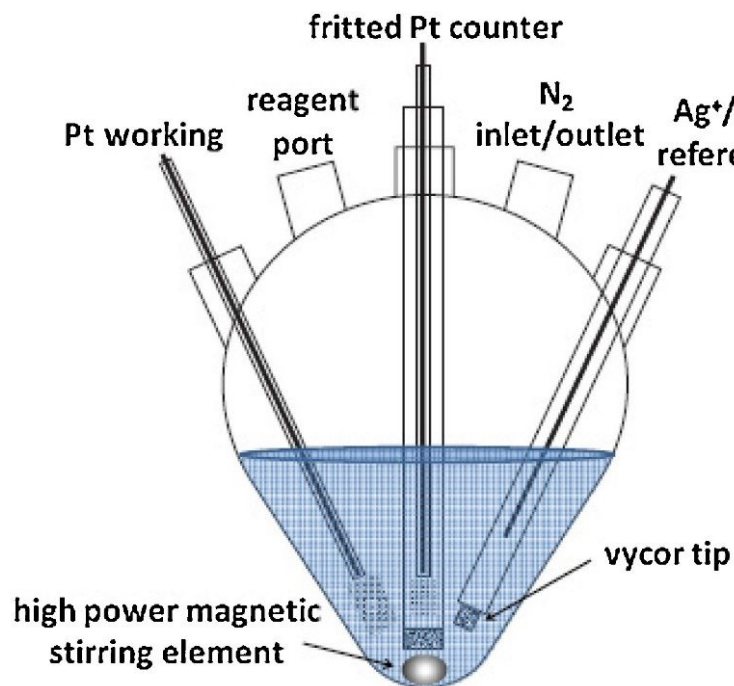
Inducing ATRP by external triggers

Electrochemical control: eATRP



Inducing ATRP by external triggers

Electrochemical control: eATRP

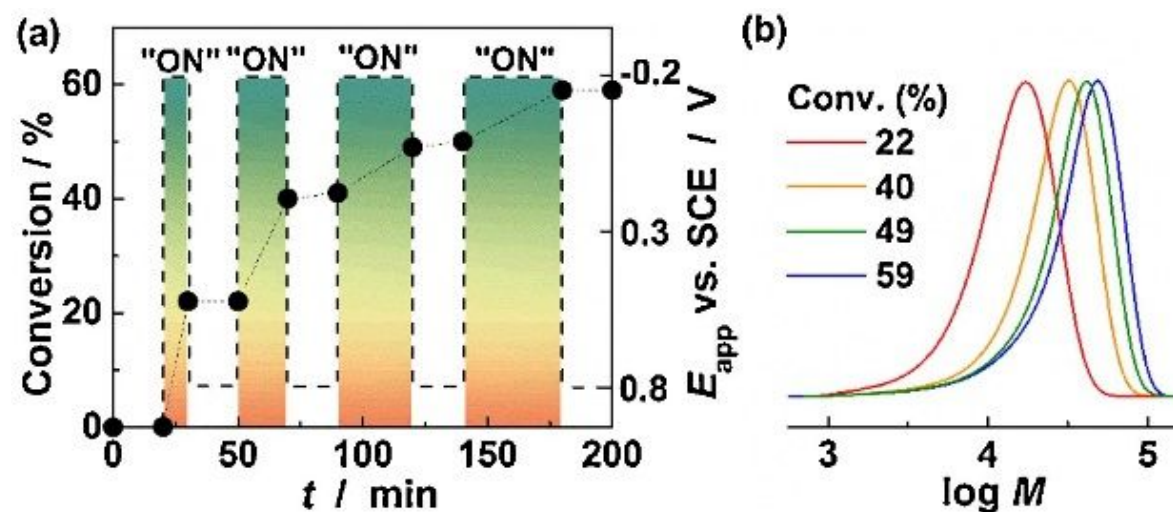
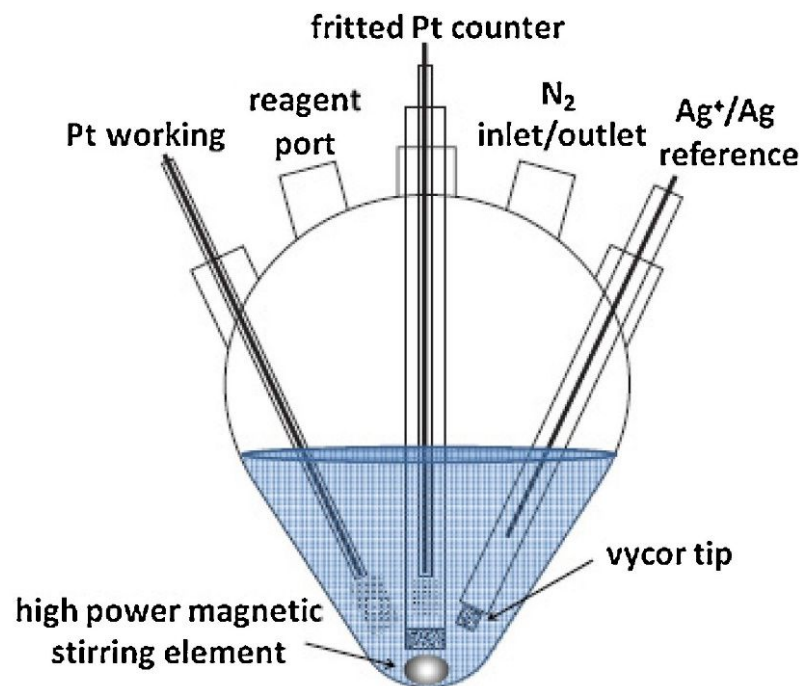


By changing the potential values (i.e., varying the [Cu^I]/[Cu^{II}] ratio), one can control the rate of polymerization.

$$E_{\text{app}} = E_{1/2} + \frac{RT}{F} \ln \frac{[\text{XCu}^{\text{II}}\text{L}]}{[\text{XCu}^{\text{I}}\text{L}]}$$

Inducing ATRP by external triggers

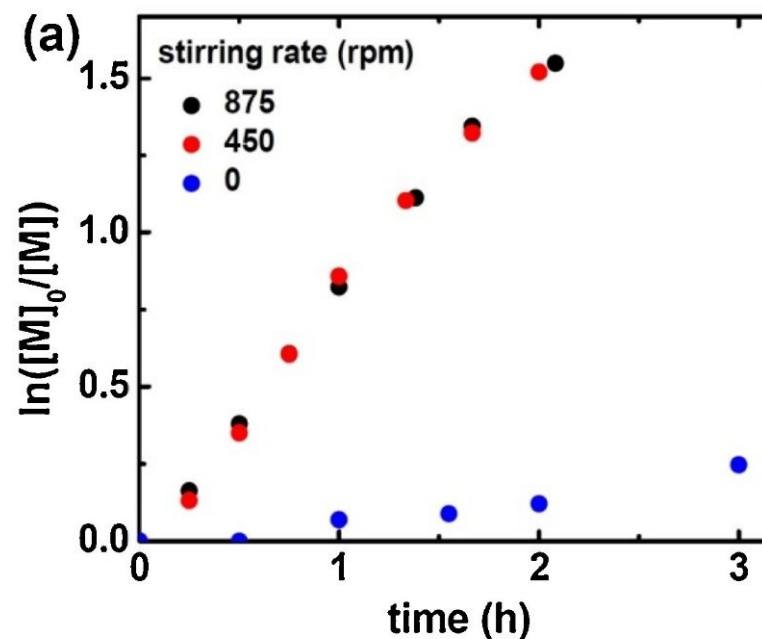
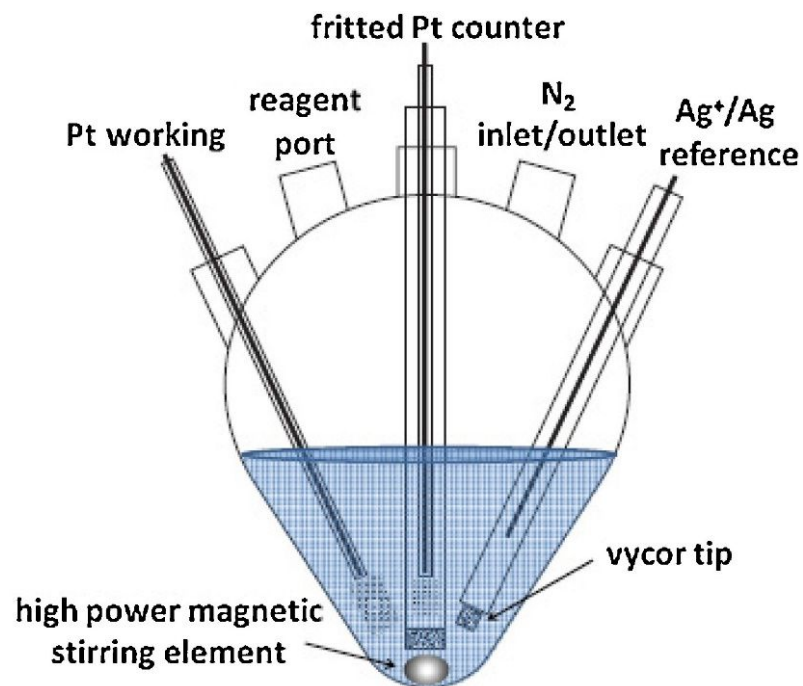
Electrochemical control: eATRP



For extreme potential values, one can turn off the polymerization.

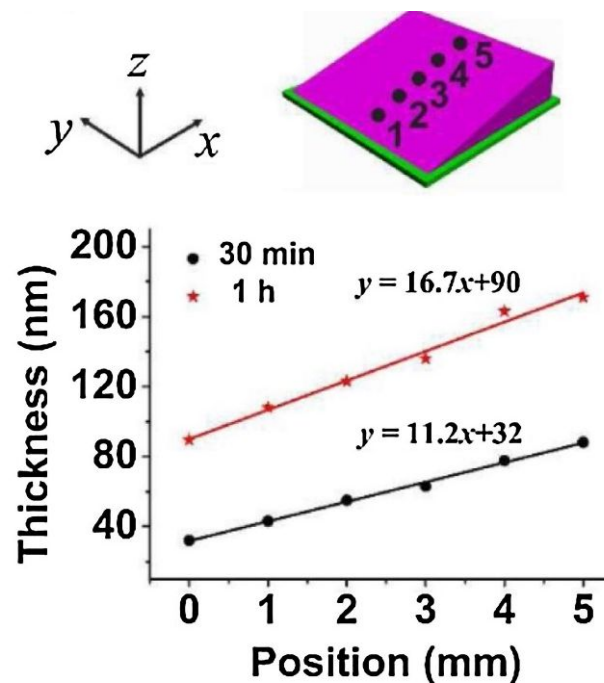
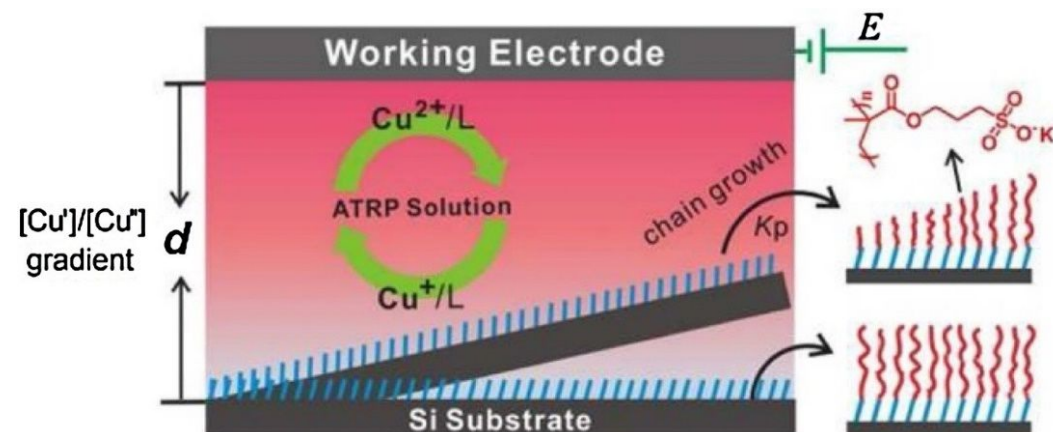
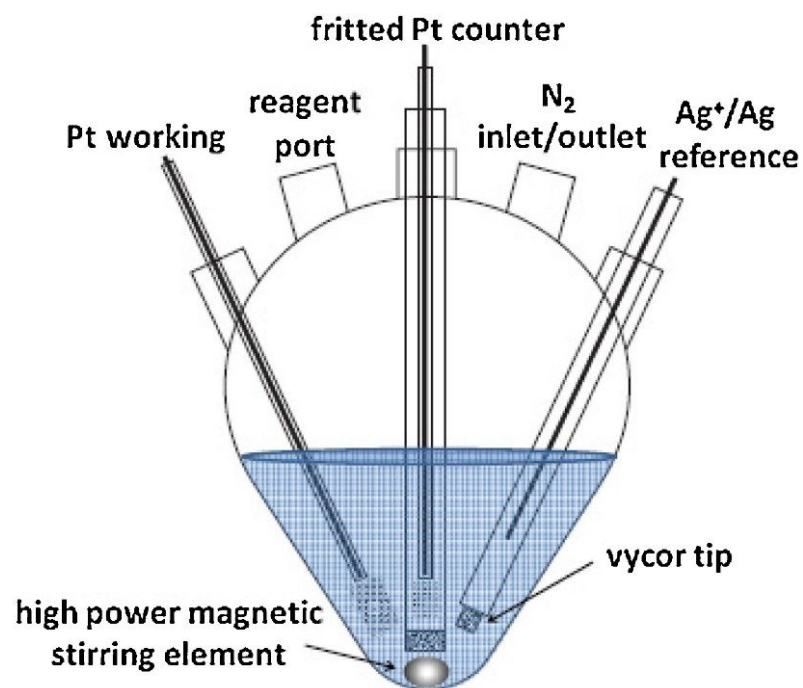
Inducing ATRP by external triggers

Electrochemical control: eATRP



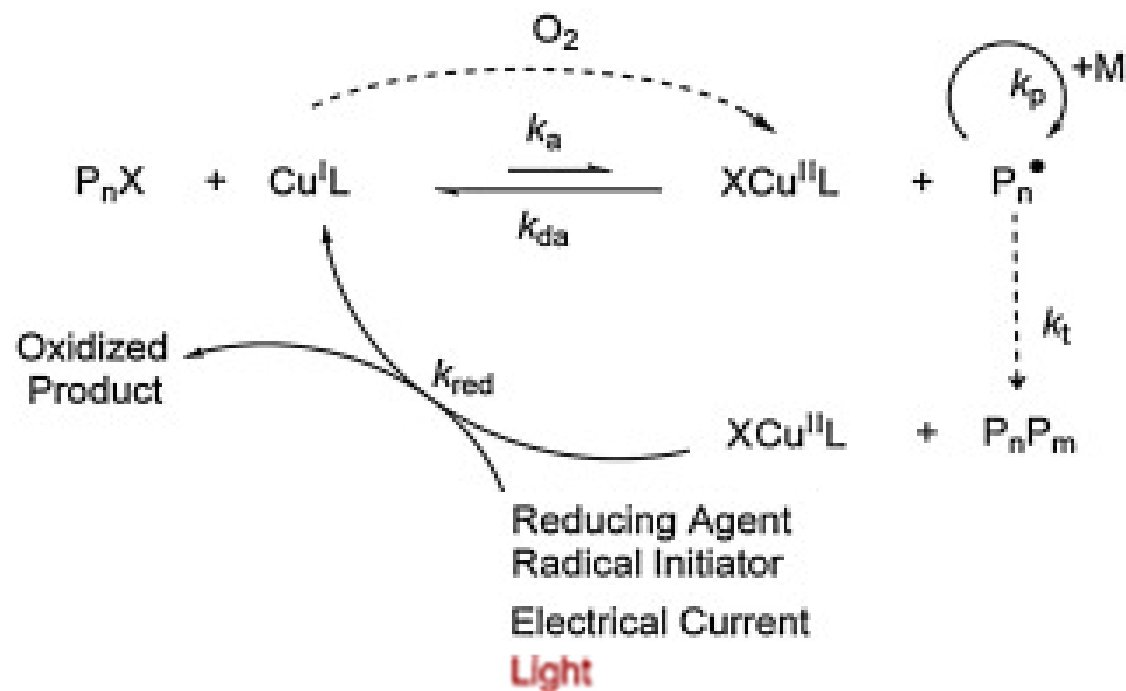
Inducing ATRP by external triggers

Electrochemical control: eATRP



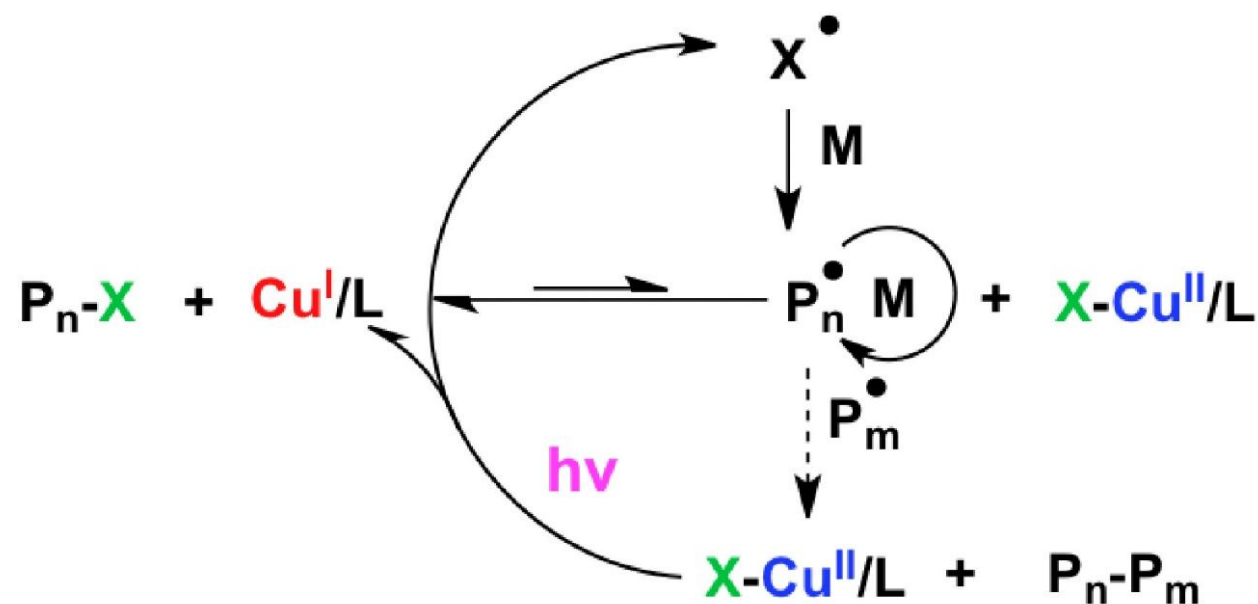
Inducing ATRP by external triggers

Photochemical control: photoATRP



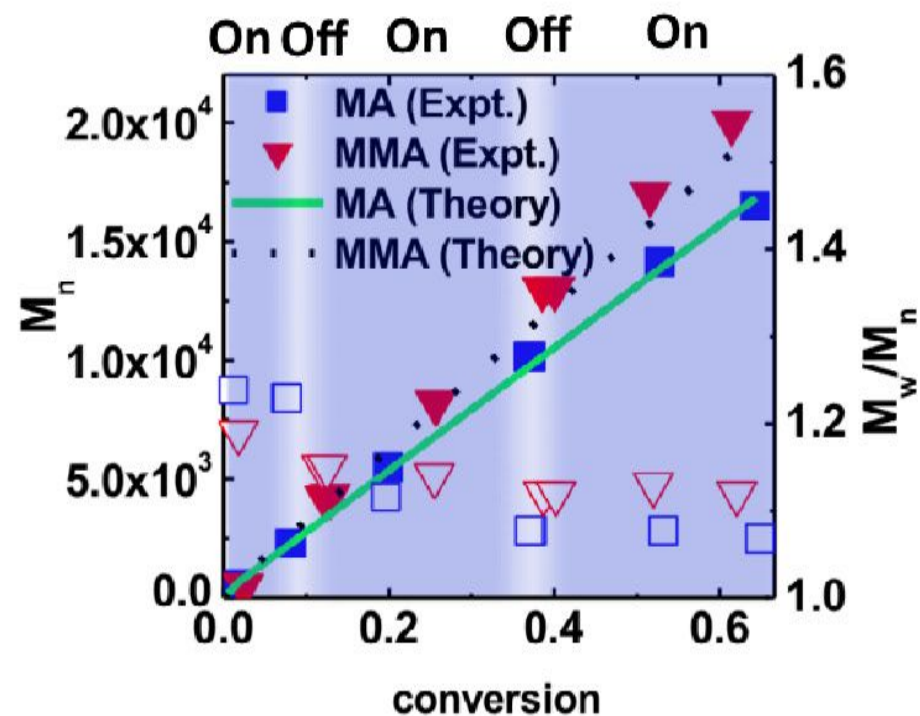
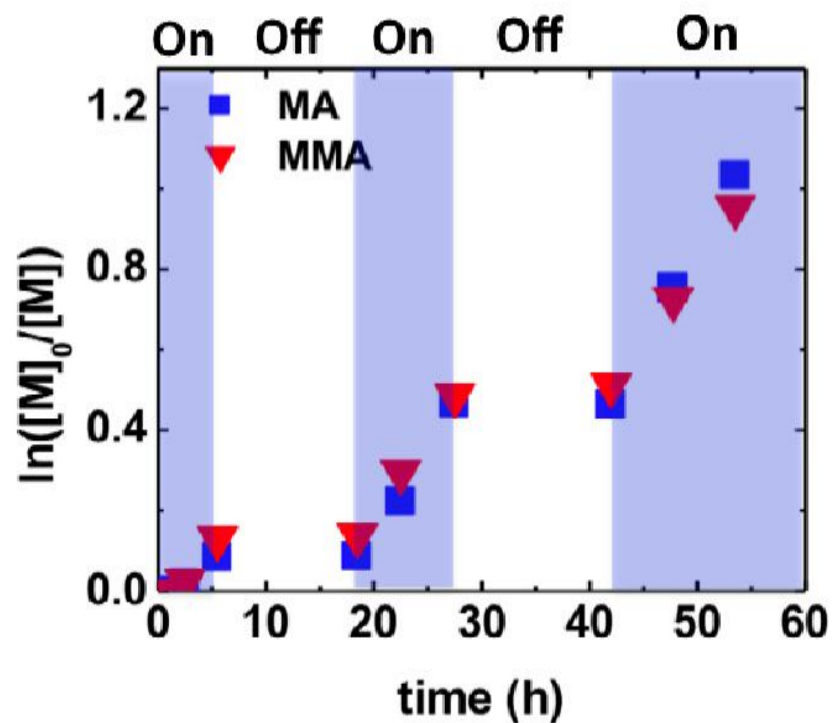
Inducing ATRP by external triggers

Photochemical control: photoATRP



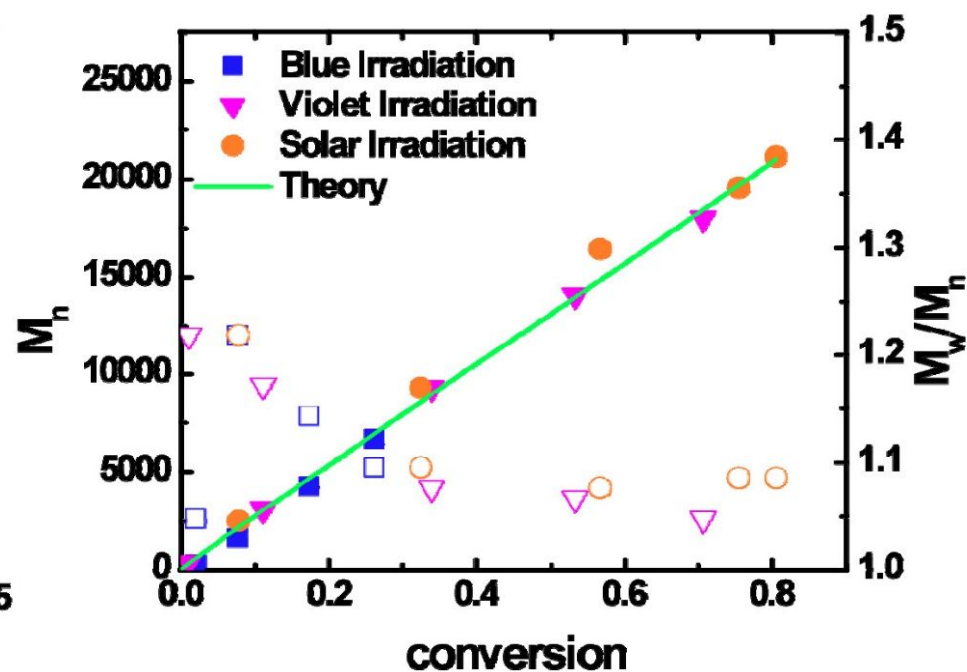
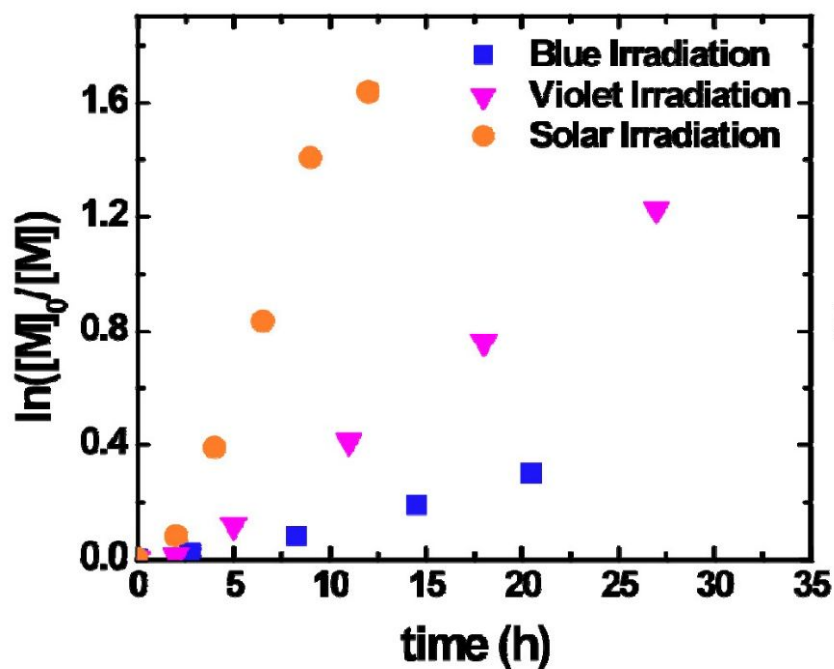
Inducing ATRP by external triggers

Photochemical control: photoATRP



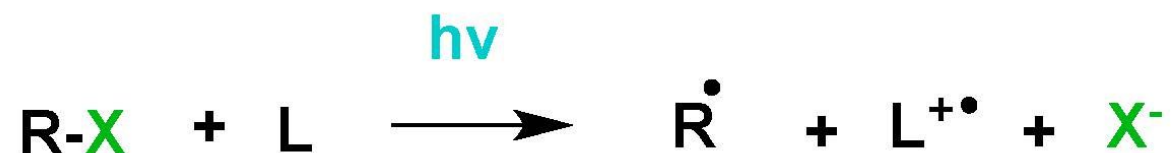
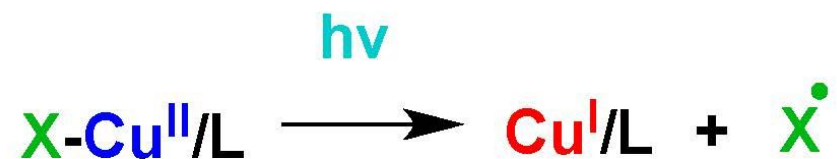
Inducing ATRP by external triggers

Photochemical control: photoATRP



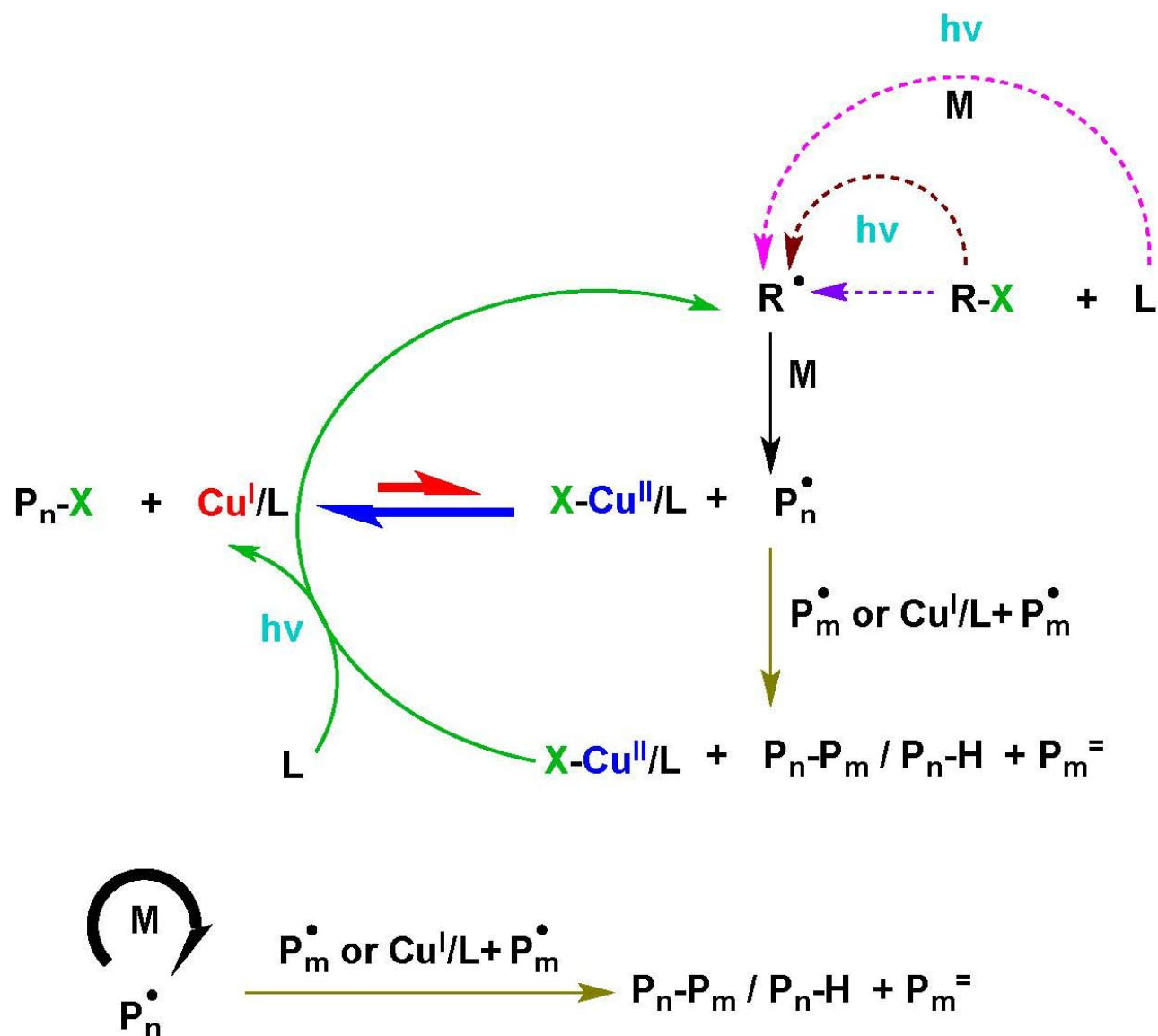
Inducing ATRP by external triggers

Photochemical control: photoATRP



Inducing ATRP by external triggers

Photochemical control: photoATRP

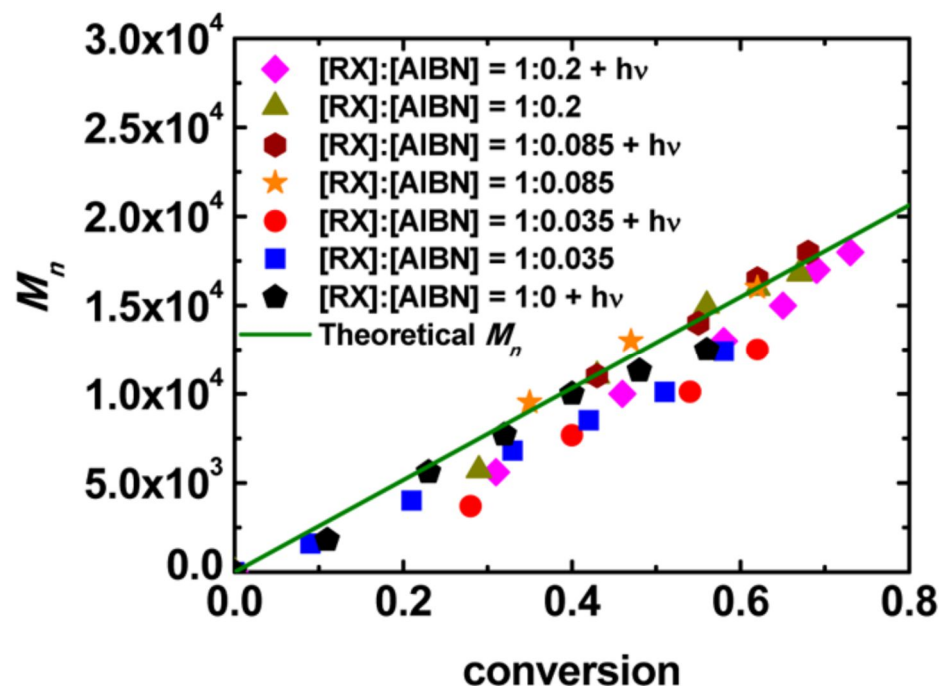
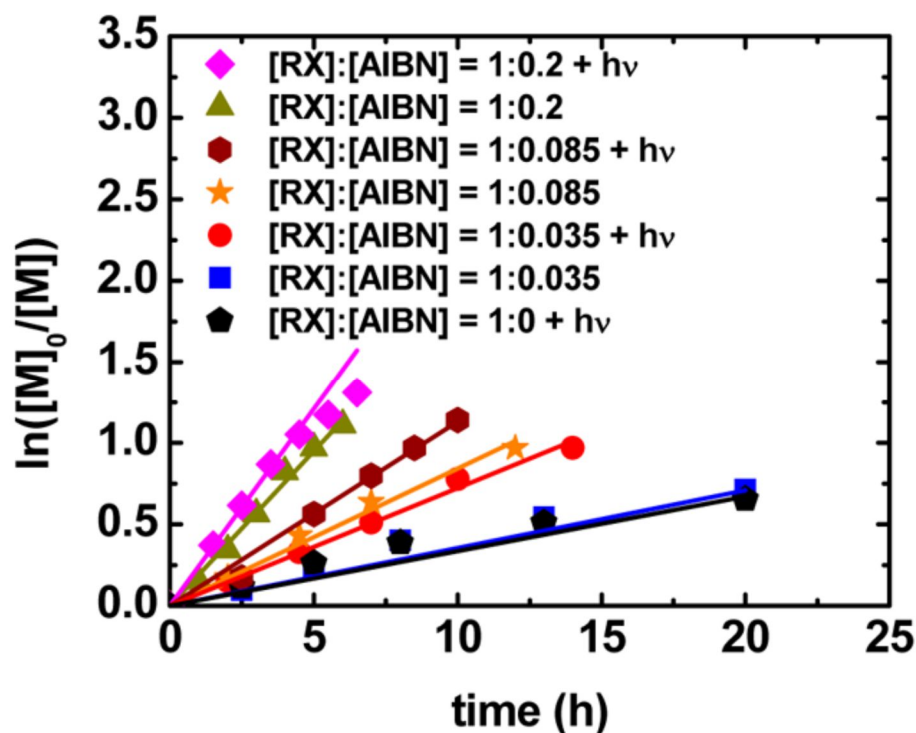


Inducing ATRP by external triggers

Photochemical control: photoATRP

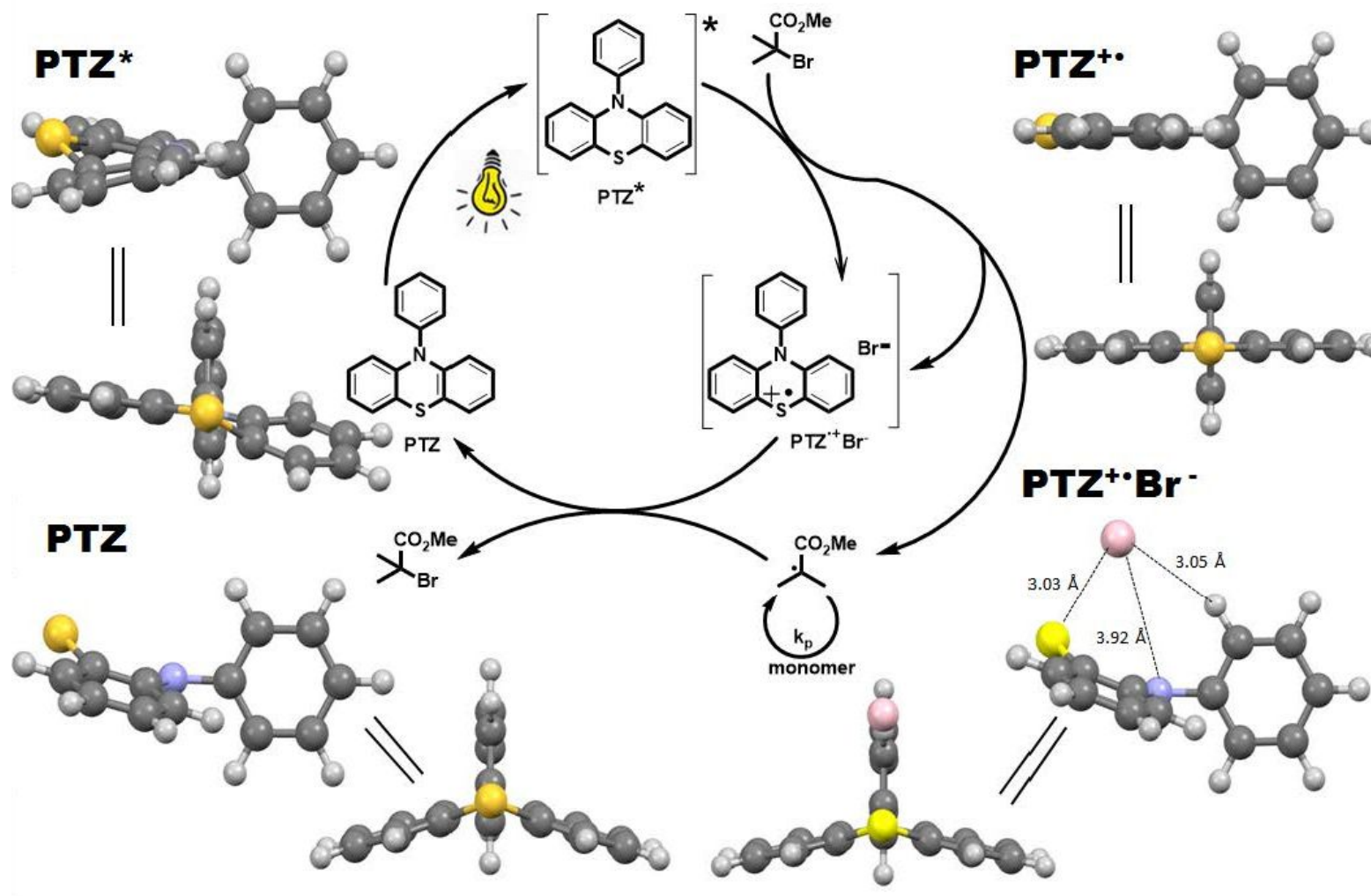
Relatively important consequence:

Performing ATRP under different lighting conditions may result in different outcomes!



Inducing ATRP by external triggers

Photochemical control: Cu-free photoATRP



Inducing ATRP by external triggers

Low Catalyst Concentrations: Summary

